Application: The Hybrid Striped Bass: Understanding Heterosis to Improve Food-Animal Ag

Benjamin Reading - bjreadin@ncsu.edu
2016 New Innovator in Food and Agriculture Research Award

Summary

ID: 0000000005
Status: Project Y3

FFAR Annual Progress Report - Y3

Completed - Dec 15 2021

Please complete this annual progress report using the spaces below the questions for your response. The report must be submitted to FFAR by the date indicated in the email notification. Disbursement of next year's funds for this grant depends on satisfactory progress on the project objectives and availability of matching funds. All progress reports must be approved by FFAR prior to disbursing FFAR funds for the following year

FFAR Annual Progress Report - Y3

Grant Information

Grant Information*

Grant ID	0000000005
Project Title	The Hybrid Striped Bass: Understanding Heterosis to Improve Food-Animal Agriculture
Reporting Period	12/01/2020-12/31/2021
Budget Period	Year 5 (Extension)
Award Program	2016 New Innovator in Food and Agriculture Research Award

Grantee Organization Information*

Organization Name	North Carolina State University
Address	2701 Sullivan Dr. Box 7514
City	Raleigh
State	NC
Zip	27695
EIN	56-6000756

Project Director/Principal Investigator Information*

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Authorized Signing Official Information*

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General Information

1. GENERAL INFORMATION

Notes on Spending

If you've experienced any deviation in the spending for your award for the last reporting period such as changes to the amount of source or matching funds, changes from your original spending plan, or carry-forward, please provide a justification below. You may enter "N/A" or "Nothing to report" if you did not experience any special circumstances.

N/A

1.1 Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the U.S., please list the city and country.*

North Carolina's 4th congressional district, Wake County, Raleigh, NC
North Carolina's 3rd congressional district, Beaufort County, Aurora, NC

1.2 How many new jobs were created by the grant during this reporting period?*

TOTAL: 6 (3.5 FTE) The below positions were funded using FFAR funds and proposed cost-share (matching funds) during year 5 of the project: 1 part-time PhD graduate student (0.5 FTE each) (L. Andersen) 1 part-time (0.5 FTE) research technician 1 part-time undergraduate researcher (Federal Work Study Student) TOTAL: 3 (1.0 FTE) The below positions all additionally contributed aspects to this FFAR project during the year 5 reporting period, although their positions were not directly funded by FFAR or FFAR cost-share (matching funds); these funds were leveraged as described below. 1 part-time PhD graduate student (0.5 FTE) 2 full-time (1.0 FTE each) research technicians Similar to the Years 1, 2, 3, and 4 Reporting Periods, this FFAR project did not fund, but it did empower the continued support of 3 full-time research technicians, 1 part-time graduate student (M. Frinsko), and 1 part-time undergraduate researcher through leveraged funds in addition to those indicated above. The technician positions were funded in part by a USDA-NIFA Hatch Multistate Project ($100,000 per year from National Research Support Project 8 (NRSP-8) National Animal Genome Research Program, National Program for Genetic Improvement and Selective Breeding for the Hybrid Striped Bass Industry) and the USDA-NIFA Southern Regional Aquaculture Center (SRAC); the part-time graduate student was funded by North Carolina Cooperative Extension. Total leveraged federal and non-federal funds during year 4 exceeded $100,000 (in addition to the originally proposed $100,000 cost-share). In addition, the National Program for Genetic Improvement and Selective Breeding for the Hybrid Striped Bass Industry currently supports farmers raising hybrid striped bass in 20 states, totaling over $56 million annually to the US economy (2018 farm gate value, without economic multipliers) and the industry directly employs a considerable number of persons nationwide (hundreds to thousands when considering the seafood support industries).

1.3 How many jobs were maintained by the grant during this reporting period?*

See 1.2 - All of the jobs listed above with few exceptions were maintained into year 5 of this FFAR project. The exceptions are as follows: of the undergraduate researchers, all had graduated except one by the end of this reporting period. The post-doctoral researcher on this project (Dr. Scott A. Salger) accepted a 9 month faculty position at Barton College (https://www.barton.edu/faculty-directory/salger-scott-a/; Wilson, NC) midway through year 1 of the project. Dr. Salger was rehired in years 2 and 3, but was not rehired again in this year (5) of the project at 0.25 FTE during the summer months of 2020 as planned due to the COVID-19 pandemic restrictions for on campus research at NC State University.

1.4 Have there been any changes to your organization’s IRS 501(c)(3) non-profit status since you were awarded the grant?*

No

Accomplishments

2. ACCOMPLISHMENTS

2.1 What were the goals/specific aims of the project for this reporting period? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets.*

The aim of the project is to characterize the paternal effects on hybrid striped bass growth performance (heterosis) and to justify the importance of broodstock striped bass currently being bred ultimately for production of hybrid striped bass in the US industry. Performance comparisons of the fish produced between and within different families will provide insight into several questions concerning breeding and growth. For example, will a large male striped bass produce large offspring with both female striped and white bass, or with only one, or with neither? This information can be applied to both aquaculture and fisheries when producing juveniles for agriculture or recreational stocking. The implications of the study can also be applied to teaching farmers how to selectively breed to produce their own juveniles, which will make the production of striped bass even more economically viable and relevant to the success of other aquaculture finfish species in the US and worldwide. Another aspect of this study is to identify non-genomic markers (metabolites) in sturgeon, a divergent fish species, to support their potential for use in predicting growth in all fishes.

The specific goals of the project are as follows:

(1) To use machine learning procedures to identify gene expression patterns that are highly predictive of growth performance in striped bass, white bass, and their hybrids;

(2) To use machine learning procedures to identify key metabolite markers associated with superior growth performance, and then demonstrate the utility of these non-genomic markers in Russian sturgeon;

(3) Ensure continued effectiveness for delivering improved broodstock to the industry.

2.2 Have any of the major goals/specific aims or milestones for this reporting period changed since the award or previous report? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals.*

There are no major changes to the research aims as originally proposed. The delays in data analysis due to technical difficulties and limitations of computational power that were reported in the previous year (4) as a result of working from home due to the novel coronavirus (SARS-CoV-2; COVID-19) were addressed and these analyses were completed in year 5.

2.3 What was accomplished under the goals/specific aims or milestones for this reporting period?*

For this reporting period describe:

1) major activities;

2) specific objectives;

3) significant results, including major findings, developments, or conclusions (both positive and negative); and

4) key outcomes or other achievements. Include a discussion of stated goals not met.

As the project progresses, the emphasis in reporting in this section should shift from reporting activities to reporting accomplishments. In the response, emphasize the significance of the findings to the scientific field. Include approaches taken to ensure robust and unbiased results.

In this reporting period, the generation and analysis of parentage and RNASeq data generated from the striped bass sampled for this project was completed. The first identification of novel pathways underlying growth heterosis in hybrid striped bass and their presentation in manuscripts that will be published in scientific journals is underway and will be completed as the graduate student working on this project (L. Andersen) prepares to complete her academic program (Summer/Fall 2022). A more detailed summary of the findings is provided in the supplemental file titled: “FFAR-2021_SupplementalFile_Sec2.3-Accomplishments.” All data, analyses results, figures, tables, and additional information generated during year 5 of this project are included as other supplemental files with this report and are described in the file inventory. This project is complete.

Objective 1: To use machine learning procedures to identify gene expression patterns that are highly predictive of growth performance in striped bass, white bass, and their hybrids;

Tool Development: The instructional machine learning pipeline developed in year 3 as a means of analyzing large datasets such as those described (e.g., 9,600,000 data points were generated and analyzed for the Hybrid Striped Bass Study 2) was used (beta-tested) to analyze transcriptome data (2,305,296 data points) generated for the Striped Bass Study.
Hybrid Striped Bass Study 1: There is nothing additional to report in year 5; this study is complete (year 2).
Hybrid Striped Bass Study 2: The study of muscle transcriptomics from a sample of the thirty half-sibling families of hybrid striped bass produced in year 1 from six white bass dams and sixteen striped bass males from differing origin (domestic, produced at the North Carolina State University Pamlico Aquaculture Field Lab, NCSU PAFL; Texas; South Carolina; Virginia; and Florida) has been completed through the stages of pathway analysis (Qiagen, Ingenuity Pathway Analysis, IPA, program. For example, five of the top pathways identified among the RNASeq data generated for this project are LXR/RXR and FXR/RXR Activation; Iron homeostasis signaling pathway; Dilated Cardiomyopathy Signaling Pathway; and Calcium Signaling. These findings will be integrated with the findings from Objective 2, Hybrid Striped Bass Study 2 (as these were the same individuals sampled) for publications and subsequent inferences to inform future research and will also be compared to the findings of Hybrid Striped Bass Study 1 and Striped Bass Study to determine what genes, pathways, and or patterns of expression may be persistent across different cohorts and hybrid versus pure strain striped bass fish.

Striped Bass Study: Microsatellite genotyping and parentage assignment of these sampled fish to their progenitors was completed during year 4, these data were integrated with the offspring metadata (e.g., measurements of traits such as weight) and used in statistical and machine learning analyses. The results of these analyses are described in more detail in the supplemental file “FFAR-2021_SupplementalFile_Sec2.3-Accomplishments.” The processing and analysis of transcriptomic data generated for this study was also completed in this year 5 of the study using CLC Genomics Workbench. All data for comparisons and analyses that have been completed to date are provided as supplemental files with this report.

Objective 2: To use machine learning procedures to identify key metabolite markers associated with superior growth performance, and then demonstrate the utility of these non-genomic markers in Russian sturgeon;

Hybrid Striped Bass Study 1: There is nothing additional to report in year 5; this study is complete (year

1).

Hybrid Striped Bass Study 2: This study is complete (year 3).

Striped Bass Study: This study is complete (year 3).

Sturgeon Study: There is nothing additional to report in year 5; this study is complete (year 2).

Objective 3: Ensure continued effectiveness for delivering improved broodstock to the industry.

The National Program for Genetic Improvement and Selective Breeding for the Hybrid Striped Bass Industry is a consortium of academic, government, and industry collaborators dedicated to improving the hybrid striped bass aquaculture through selective breeding and domestication and works alongside the NOAA Sea Grant StriperHub. These fish (broodstock) are produced annually at the NCSU PAFL, presently the sole world source of domesticated striped bass. In 2017, 2018, 2019, 2020, and 2021 approximately 116, 125, 140, 100, 108 respectively, different families of domestic striped bass were produced along with over 50 crosses of domestic white bass in years 2018, 2019, and 2021. Over 400 adult domestic, genetically improved male striped bass were disseminated to commercial hybrid striped bass farmers in North Carolina and South Carolina from 2017-2021 and broodstock animals also were distributed throughout the Midwest from the USDA ARS Harry K. Dupree Stuttgart National Aquaculture Research Center. Hybrid striped bass farming is the fourth largest form of finfish aquaculture in the US, behind catfish, salmon, and trout with an annual farm-gate value of over $54 million. More than 90% of the hybrid striped bass raised in the US in 2017-2021 were produced using these domesticated broodstock. In addition, millions of striped bass, white bass, and hybrid striped bass fry and hundreds of thousands of juvenile domestic fish were transferred into the industry or provided to other research groups in year 5. Domesticated white bass were used to produce hybrid striped bass at the North Carolina Watha State Fish Hatchery, which were used to stock freshwater reservoirs to support recreational fisheries throughout the state from 2017 to 2021 (Dingell-Johnson, Sport Fish Restoration Act). The estimated U.S. economic impact of the striped bass breeding program during this FFAR project (from 2017-2021) is $200 million based only on farm gate value of the seafood products (>300 ROI of the funds).

2.4 Describe challenges or delays encountered during the reporting period and actions or plans to resolve them. Only describe significant challenges that may impede the research and emphasize their resolution.*

There is nothing to report.

2.5 Have there been any changes in scientific approach or reasons for change? If so, what are the changes? Remember, changes to the approved scientific approach must be pre-approved by FFAR.*

No significant changes to the scientific approach are proposed.

2.6 What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals.*

In year 5 of the project the one PhD graduate student funded directly by FFAR (L. Andersen) was accepted to be a member of the Association for International Agriculture and Rural Development (AIARD) Future Leaders Forum (FLF). The FLF program grants members an opportunity to network with various global leaders in agriculture. L. Andersen is serving on the AIARD Communications Committee and is participating in the AIARD FLF mentorship program under the guidance of Dr. NseAbasi Etim (Department of Animal Science, Akwa Ibom State University, Nigeria). To learn more about the AIARD FLF program, please visit: http://www.aiard.org/2021-future-leaders-forum.html. L. Andersen was also selected to be a Center for Environmental Farming Systems (CEFS) Graduate Fellow. The CEFS Graduate Fellows Program provides financial support and recognition for the future leaders, researchers and contributors in sustainable agriculture and local food systems while they pursue academic research. To learn more about the CEFS Graduate Fellows program, please visit: https://cefs.ncsu.edu/academics-and-education/graduate-fellows/.

L. Andersen was also able to gain immersive training and experience in teaching and the development of instructional materials. L. Andersen served as the Teaching Assistant for the Genome Engineering: CRISPR Technology course (BIT 495/595) offered at NC State University and is the lead author on a teaching manuscript that is currently in preparation and describes the teaching strategy and developed materials used to instruct this course online during the in-person school closures (due to SARS-CoV-2; COVID-19). L. Andersen took the lead to develop these materials and instructed the course virtually during the Spring 2020, Fall 2020, and Spring 2021 semesters. In Fall 2021 L. Andersen also served as a Teaching Assistant for Biology of Fishes (AEC 441) with PI Reading as instructor. Additionally, L. Andersen was accepted into and completed the Preparing the Professoriate (PTP) program offered by NC State

University that began in September 2020 of year 4. The PTP program was established in 1993 and is a nationally recognized, selective program (30 of over 200 applicants are accepted) designed to give exceptional doctoral students and postdoctoral scholars an immersive mentoring, teaching, and future faculty preparation experience (https://grad.ncsu.edu/professional-development/career-support/ptp/). L. Andersen also participated in field activities for the National Program for Genetic Improvement and Selective Breeding for the Hybrid Striped Bass Industry and the recently created StriperHub network (PI Reading as Program Coordinator), including communicating with farmers and stakeholders, and networking with aquaculture industry and agriculture extension persons, and state and local government (see:
https://ncseagrant.ncsu.edu/news/2020/12/breakthroughs-position-striped-bass-for-commercial-success/?fbclid=IwAR1iRN4sZawIecwKPP-pm8OFeZFFnQRN-5JgtyiB_Qez_00l4js3IDWewYs
https://cals.ncsu.edu/applied-ecology/news/striperhub-will-address-seafood-deficit/).

2.7 Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail.*

The PI mentors 1 undergraduate researcher who has contributed to objectives in year 5. Two graduate students (2 PhD) also were involved with this FFAR project.

The PI served as a Faculty Mentor for high school students in the CAALS 3-D program during years 1, 2, and 3 of this FFAR project. This program is held in partnership with North Carolina School of Science and Mathematics, Triton High School, Wake Young Women's Leadership Academy, Enloe High School, Calvary Day School, Enloe Magnet High School, and North Wake College & Career Academy. The CAALS 3-D mission is to increase awareness and interest of high school students from under-represented groups (URM) in STEM career fields within the food, agricultural, environmental and life sciences. During years 4 and 5 of the project, this summer program was unfortunately cancelled due to COVID-19.

The PI taught the following courses: Machine Learning Approaches in Biological Sciences (AEC510), Biology of Fishes (AEC441), and Biology of Fishes Laboratory (AEC442). The machine learning course was developed by the PI based on the unique methods used in this FFAR project. The course was taught in 2018 (year 2), 2019 (year 3), 2020 (year 4), and 2021 (year 5). This is now a permanent course offering (AEC510) and over 60 students have been trained in the novel machine learning curriculum. In 2022, an undergraduate course offering AEC410 will be developed.

Striped bass were provided to the Aquaculture Technology Program at Carteret Community College (Morehead City, NC) for education and aquaculture demonstration each year from 2017 to 2021. The PI gave guest lectures on the importance of hybrid striped bass, aquaculture, and global food security, and also provided demonstrations, workshops, and outreach/extension activities in particular through the new StriperHub that was established in 2020 with NOAA and NC Sea Grant.

2.8. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products.*

Visitors and/or beneficiaries involved in workshops, meetings, tours, and other outreach activities at the National Program for Genetic Improvement and Selective Breeding for the Hybrid Striped Bass Industry at the NCSU PAFL: 53 persons in 2016 (27 for commercial or business reasons, 23 for educational or research reasons and 3 for North Carolina agriculture extension), 33 in 2017 (5 for commercial or business reasons, 26 for educational or research reasons and 2 for North Carolina agriculture extension), 40 in 2018 (7 for commercial or business reasons, 15 for educational or research reasons and 17 for North Carolina agriculture extension), and 21 in 2019 (5 for commercial or business reasons, 13 for educational or research reasons and 3 for North Carolina agriculture extension), and over a hundred persons in year 4 (2020). Most public tours were cancelled in 2021 due to COVID-19 and limited to a few dozen persons. These persons included North Carolina Agriculture Extension and North Carolina Department of Agriculture and Consumer Services personnel, commercial fish farmers, professors and researchers, graduate and undergraduate students, researchers from the USDA ARS and NC Wildlife Resources Commission (NC WRC), NC Division of Marine Fisheries (NC DMF), and Boy Scouts of America. Additionally, aquaculture extension tours conducted at the Lake Wheeler Field Laboratory Fish Barn (Raleigh, NC) included approximately 250 visitors each year in 2018, 2019, and 2020, however these tours were limited in 2021 due to COVID-19.

2.9. What do you plan to do during the next reporting period to accomplish the goals of the approved project? Briefly describe what you plan to do during the next reporting period to accomplish the project goals and objectives. Discuss efforts to ensure the approach is scientifically rigorous and results are robust and unbiased. Remember that significant changes to the approved goals and objectives and project scope require prior FFAR approval. Include any important modification to the original goals and provide justification for the change.*

This project is complete. We are working on final writing of the manuscripts for submission to publication and anticipate these results will be published in 2022.

Information Products

3. INFORMATION PRODUCTS

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during this reporting period resulting directly from the FFAR award.*

Scholarly publications = 4 (2 published and 2 submitted for publication)

Workshop summaries = 4

Lectures and seminars = 3

Student presentations = 3

Scientific presentations = 2

Industry/government forums = 2

Invention disclosures = 0

Press releases and publicity = 3

Videos and Images = 1 instructional videos, hundreds of additional videos and images posted on https://www.facebook.com/stripedbassgenome/

Curricula = 1 course developed (AEC 510 Machine Learning Approaches in Biological Sciences). Twelve (12) graduate students were trained in 2021 and a total of 61 students have been trained in this course while it was under development during this FFAR project.

The research has been disseminated in print, oral, and digital formats to several hundreds of audience members in the scientific community and public venues and online with hundreds of accessions daily (many thousands of persons reached through the internet).

3.2 Please select all types of information products produced during this reporting period resulting directly from the FFAR award.*

Responses Selected:

Scholarly publications
Workshop summaries
Videos
Images
Curricula
Other: Press releases and publicity

3.3. Please provide a list of citations for the information products produced during this period.*

The following scientific presentations have been delivered (or abstracts scheduled for presentation) and either reference the FFAR for funding support, detail the research and/or the importance of domestic aquaculture, or involve the PI as part of this FFAR project († denotes student author):

Workshops

14. Reading, B.J., Lopez, F., Herbst, E., Borski, R.J., Berlinsky, D.L., Bolton, G., Nash, B., Ciaramella, M., Rawles, S., Turano, M., Parker, M., Cerino, D., Brown, T., Fuller, S.A., Abernathy, J., Pigg, S., Booker, M., Won, Eugene, and Chambers, M. 2021. NOAA Sea Grant Striper Hub. NC Aquaculture Development Conference. March 18. Remote Presentation, Morehead City, North Carolina, USA.

13. Reading, B.J., Daniels, H.V., Borski, R.J., Hinshaw, J., Chouljenko, A., Bolton, G., and Hall, S. 2021. University Freshwater Research Updates. NC Aquaculture Development Conference. March 19. Remote Presentation, Morehead City, North Carolina, USA.

12. Herbst, E., Dominguez, C. (chef), Nash, B., and Reading, B.J. 2021. StriperHub: Farm Raised Striped Bass / Sauteed Garlic Basil-Butter Striped Bass. The 35th Annual NC Seafood Festival, Cooking With the Chefs. October 3. Morehead City, North Carolina, USA.

11. Reading, B.J., Onley, A., Dominguez, C. (chef), and Nash, B.. 2021. StriperHub: Farm Raised Striped Bass / Sauteed Garlic Basil-Butter Striped Bass. The 35th Annual NC Seafood Festival, Cooking With the Chefs. October 2. Morehead City, North Carolina, USA.

Lectures and Seminars

10. Andersen, L.K.† 2021. Machine Learning Analysis of Genes Underlying Growth Performance in Aquacultured Fish (Genus Morone). Association for International Agriculture and Rural Development (AIARD) Fireside Chat. November 18. Remote Presentation, Raleigh, North Carolina, USA.
9. Reading, B.J. (Panelist). 2021. Junior Faculty Forum. North Carolina State University, Office of Research and Innovation, Research Leadership Academy (RLA). September 28. Remote Presentation, Raleigh, North Carolina, USA.
8. Reading, B.J.. 2021. StriperHub: Striped Bass Aquaculture. Coastal Resilience and Sustainability. North Carolina State University, Coastal Resilience and Sustainability Initiative. September 22. Remote Presentation, Raleigh, North Carolina, USA.

Student Presentations (undergraduate and high school)
7. Faizi, I.† and Hansen, M.†. 2021. Ossetra Caviar Microbiome. North Carolina State University Summer Undergraduate Research & Creativity Symposium. July 29. Remote Presentation, Raleigh, North Carolina, USA.
6. Pereira, S.†. 2021. Developing a Clearing and Staining Protocol to Enhance Fisheries Education. North Carolina State University, Department of Applied Ecology Minor Symposium. April 30. Remote Presentation, Raleigh, North Carolina, USA.
5. Pereira, S.†. 2021. Developing a Clearing and Staining Protocol to Enhance Fisheries Education. North Carolina State University Spring Undergraduate Research & Creativity Symposium. April 22. Remote Presentation, Raleigh, North Carolina, USA.

Scientific Presentations
4. Reading, B.J., Andersen, L.K.†, Abernathy, J., Berlinsky, D.L., Bolton, G., Borski, R.J., Cerino, D., Ciaramella, M., Frinsko, M.O.†, Fuller, S.A., Gabel, S., Green, B.W., Herbst, E., Hopper, M., Kenter. L.K.†, Lopez, F., McGinty, A.S., Nash, B., Parker, M., and Rawles, S. 2022. StriperHub: Striped Bass (Morone saxatilis) Aquaculture. Aquaculture Triennial 2022 (World Aquaculture Society). February 28-March 4. San Diego, California, USA.
3. Andersen, L.K.†, Valdivia-Granda, W.A., and Reading, B.J. 2022. Expression Profile Analysis Via Machine Learning to Identify Critical Genes Determinant of Hybrid Striped Bass Morone chrysops X M. saxatilis Growth. Aquaculture Triennial 2022 (World Aquaculture Society). February 28-March 4. San Diego, California, USA.

Industry/Government Forums
2. Reading, B., Borski, R., Nash, B., Lopez, F., Herbst, E., Cerino, D., Berlinsky, D., Ciaramella, M., Parker, M., Snyder, B., and White, S. 2021. Establishing the Sea Grant Striped Bass Aquaculture Hub (StriperHub): Commercialization, Economics, and Marketing. Sea Grant Aquaculture Research Symposia.

November 1. Remote Presentation, Silver Spring, Maryland, USA.

1. Bashura, J., Burke, M., Mikulec Jr., J., Owens, T., Reading, B.J., Richt, J.A., Spencer, D., Valdivia-Granda, W.:, Weekes, J., Wright, D., Wittrock, M., and Lapitan, R. 2021. Threats to Food and Agriculture Resources (TFAR). Public-Private Analytic Exchange (AEP) Virtual Concluding Summit, United States Department of Homeland Security (DHS) Office of Intelligence and Analysis (I&A) and the Office of the Director of National Intelligence (ODNI). August 25. Remote Presentation, Washington, DC, USA.

3.4. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the reporting period resulting directly from the FFAR award?*

Yes

3.4b. If yes, please provide citation.

The following manuscripts were published and directly reference the FFAR for funding support (collaborations with USDA ARS and USDHS I&A):

2. Bashura, J., Burke, M., Lapitan, R., Mikulec Jr., J., Owens, T., Reading, B.J.*, Richt, J.A., Spencer, D., Valdivia-Granda, W.*, Weekes, J., Wittrock, M., and Wright, D. 2021. Threats to Food and Agricultural Resources. United States Department of Homeland Security, Office of Intelligence and Analysis, Analytic Exchange Program. 95 pp. *Authors wrote the first draft and initial revision of this manuscript (authors are otherwise listed in alphabetical order).
https://www.dhs.gov/sites/default/files/publications/threats_to_food_and_agriculture_resources.pdf

1. Prior, B.S., Lange. M.D., Salger, S.A., Reading, B.J., Peatman, E., and Beck, B.H. 2021. The effect of piscidin antimicrobial peptides on the formation of Gram-negative bacterial biofilms. Journal of Fish Diseases https://doi.org/10.1111/jfd.13540.

The following manuscripts were published and reference data/findings reported from this FFAR project and/or use machine learning methods developed in this FFAR project:

The following manuscripts are in review/press/preparation and directly reference the FFAR for funding support or use machine learning methods developed in this FFAR project:

4. Giacomini, J.J.†, Adler, L.S., Reading, B.J., and Irwin, R.E. In review. Differential bumble bee gene expression associated with pathogen infection and pollen diet. BMC Genomics.
3. Damiano, M.D.†, Cao, J., Reading, B.J. In review. Predicting multi-species marine fish distribution using support vector machines: a case study from the Southeast USA.
2. Rajab, S.†, Andersen, L.K.†, Kenter, L.†, Berlinsky, D.L., Borski, R.J., McGinty, A.S., Ashwell, C.M., Ferket, P., Daniels, H.V., and Reading, B.J. In preparation. Combinatorial metabolomic and transcriptomic analysis of muscle growth in sunshine hybrid striped bass (female white bass Morone chrysops x male striped bass M. saxatilis).
1. Andersen, L.K.†, Sheoran, D., Carter, R.E., Valdivia-Granda, W.A., and Reading, B.J. In preparation. Expression profile analysis via machine learning to identify genes determinant of hybrid striped bass (Morone chrysops x M. saxatilis) growth.

3.5. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.*

1. Founding Member of the Striped Bass Genome Community (StriperHub), a FaceBook web page designed and updated to provide information, videos, and pictures about current research, news, education, and outreach. On average, hundreds of people were served daily activity from this page during the last reporting period with a daily high in the several thousands.
Follow us on Facebook https://www.facebook.com/stripedbassgenome/

2. Coordinator of the Striped Bass and White Bass Genome Sequencing Projects. We provide a publicly accessible, online database for the striped bass genome sequence assembly.
Homepage: http://appliedecology.cals.ncsu.edu/striped-bass-genome-project/
Transcriptome Database: https://www.animalgenome.org/aquaculture/s.bass/

3. Striped bass is a priority species for the USDA National Research Support Project 8 (National Animal Genome Research Program). The PI is the NRSP-8 Striped Bass Species Coordinator, the NRSP-8 National Aquaculture Co-Coordinator, NOAA Sea Grant StriperHUB Program Coordinator. More information about striped bass genome resources and other important agriculture food animals are provided at:
https://www.animalgenome.org/

4. A PhD Student (L. Andersen) funded through this project maintains a website describing her research and teaching efforts as a professional portfolio and to promote ongoing research: http://lkandersen.com

Other Relevant Press Releases and Research Publicity:

3. Pure-strain striped bass: An opportunity waiting to be tapped. Aquaculture North America. October 13, 2021.
https://www.aquaculturenorthamerica.com/pure-strain-striped-bass-an-opportunity-waiting-to-be-tapped/?fbclid=IwAR3BC_qUsDyGn142Cs5v3c6r1svof0cjJ0ayor6RlI5nnIVn0GfhIcwxBO8

2. Science, Magic, and More Fill the Spring 2021 Issue of Coastwatch. North Carolina Sea Grant News. March 24, 2021.
https://ncseagrant.ncsu.edu/news/2021/03/science-magic-and-more-fill-the-spring-2021-issue-of-coastwatch/?fbclid=IwAR2fMUOFjswY7gpheP4dPj1yWmptoPUIr4D9QLTkEzc_Uhc6EVG7Mngbej0
1. Magic at 64.4 Degrees: Science, Serendipity, and Farmed Striped Bass. North Carolina Sea Grant

Coastwatch. Spring 2021. (Featured on Cover)

https://ncseagrant.ncsu.edu/coastwatch/current-issue/spring-2021/magic-at-64-4-degrees/?fbclid=IwAR19RHw4K3FNx2lUwp4hsra6JwDPw3YmuYwbDB7TWIiPW4vNfJF5piyFF9I

Instructional Videos:

The following video was recorded and posted for the Biology of Fishes course and K-12 outreach during the spring semester of this reporting period

1. Striped Bass Dissection and Q&A: This is a recording of striped bass dissection that was first streamed on February 16th, 2021. Hosted by Ph.D. candidate, Linnea Andersen, Prof. Ben Reading, and Michelle Jewell of NC State's Department of Applied Ecology.
https://www.youtube.com/watch?v=S85IwuOyzbE

3.6. Have inventions, patent applications, and/or licenses resulted from the award during this reporting period?*

Yes

3.6b. If yes, indicate the invention, patent application(s), and/or license(s).

We submitted an invention disclosure during year 4 (2020) of this FFAR project through the NC State University Office of Research Commercialization that describes the machine learning analysis pipeline developed for the analysis of data generated from this project.

3.7. Are any of the information products produced during this reporting period confidential, proprietary, or subject to special license agreements?*

No

3.8. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?*

Data are archived internally at NC State University as well as hosted online through the NRSP-8 National Animal Genome Project and NC State University Applied Ecology websites (the PI is the National Aquaculture Coordinator for the NRSP-8). We have submitted data to the National Center for Biotechnology Information (NCBI) for archiving and it is publicly available. These data are: the SB reference genome (NCSU_SB_2.0; accession number: GCA_004916995.1); the WB reference genome (DOM_MoChry_2.0; accession number: GCA_019097615.1); and transcriptomics data from the seventy-two SB sampled for this project (BioSamples published under 72 unique accession numbers from SAMN12248753-SAMN12248824)

NC Area Agriculture Extension Agents are involved with and aware of the project and have disseminated information about it to stakeholders in the state.

The PI is a member of the USDA NIFA Southern Regional Aquaculture Center (SRAC) and the project is known through that regional extension network, which includes academics as well as commercial producers and state extension agents in the southeast U.S.

The PI is a member of the North Carolina State University Animal Food and Nutrition Consortium and the Food Animal Agriculture Initiative, and the project is known through that regional network, which includes large global corporate members such as BASF, Syngenta, Novozymes, and Premex.

The PI is Program Coordinator for the NOAA Sea Grant StriperHUB, a nexus to commercialize striped bass aquaculture, and Co-Coordinator of the National Program for Genetic Improvement and Selective Breeding for the Hybrid Striped Bass Industry; information is disseminated to academic, government, and industry partners through these extension programs.

The PI also served as a Subject Matter Expert (SME, threats and vulnerabilities related to seafood and aquatic resources) to the US Department of Homeland Security Office of Intelligence and Analysis Analytic Exchange Program (AEP) during year 5 and contributed to the Threats to Food And Agricultural Resources national intelligence report issued to the Office of the Director of National Intelligence (ODNI). This report was presented at the annual AEP Intelligence Summit.

Data Management

4. DATA MANAGEMENT

FFAR is developing a more nuanced Data Management Policy. As part of this effort, we are introducing new questions into the Annual Reports on Data. All questions on this page are required to have a response. In cases where your answer may be repeated, you are welcome to copy-paste relevant information. Thank you for your patience as we update our reporting portal.

4.1. During this period, did the project generate any data?*

Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries).

Yes

4.1b. Please list the data generated in this reporting period.

The project did generate data. The data generated during this year (year 5) of the project are provided in a file folder (Year_5-EXTENSION_Data_FFARAnnualProgress_ID-43087.zip). All files and folders listed below are backed up on external hard drives in addition to being in this folder submitted to FFAR. A copy of this list is included in the zipped folder mentioned above and named “FFAR_Report_File_Inventory_Year5-EXTENSION_2021.docx.”

4.1c. If you list multiple data sets, are these data sets related?

Yes

4.1d. If so, please provide a short description of how they are related.

The data sets in the files and folders listed below are related as follows:

Multiple Objectives/Studies

File no. 1 is a document describing, in greater detail, the progress that has been made during the final year, year 5, of this FFAR Project (specifically, Objective 1 Hybrid Striped Bass Study 2 and Striped Bass Study).

Objective 1, Hybrid Striped Bass Study 2:

Folder no. 11 contains output from QIAGEN IPA (Ingenuity Pathway Analysis) performed using the twenty-nine (29) genes that the 832 motif-fingerprints identified as being important in determining Grade (Top Grade, Runts), Growth (Large, Intermediate, Small), and Strain (Domestic, Florida, Virginia, South Carolina, & Texas) of the hybrid striped bass sampled for the Hybrid Striped Bass Study 2 of this project.

Objective 1, Striped Bass Study:

File no. 2 is an excel file with updated metadata and sample coding information for the striped bass sampled and included in analyses for Objectives 1 and 2, Striped Bass Study. This file was also used as metadata input in CLC Genomics Workbench to conduct differential expression analysis.

File no. 3 is an .XLSX file containing all of the read data (RPKM) of gene transcripts identified (32,018 transcripts total) among the striped bass reference genome and the Objective 1, Striped Bass Study RNASeq data collected from seventy-two (72) individual striped bass. These values were calculated in CLC Genomics Workbench and are referred to as “Transcript expression (TE)” data. Data is associated with RNASeq sample names and their group identifiers (Grade, Grade Group, Dam, Sire, Sire Group) as according to the metadata file (File no. 1). Subsequent sheets contain only the RNASeq sample name and specific group identifiers. Copies were made of the group-specific sheets and used as input into R to generate transposed CSV files that could be converted into ARFF files for WEKA analysis.

Folder no. 4 contains the output of RNASeq read data for samples 3-9, 39-72 (data for samples 1, 2, and 10-38 was provided in year 4 of this project, 2020) in two separate zipped folders: CLC_RNASeqAnalysis_FFAR-SB-3-9_39-52.zip (sub-folder 3a) and CLC_RNASeqAnalysis_FFAR-SB-53-72.zip (sub-folder 3b). These data were generated using the RNASeq Analysis tool in the Transcriptomics Analysis toolbox within CLC Genomics Workbench. This approach is based off of that taken by Mortazavi et al. (2008), full citation: Mortazavi, A., Williams, B.A., McCue, K., Schaeffer, L. and Wold, B., 2008. Mapping and quantifying mammalian transcriptomes by RNA-Seq. Nature methods, 5(7), pp.621-628. The transcript expression (TE) files that were used for machine learning and other analyses are included here, among other files.

Folder no. 5 contains an additional copy of the metadata for Objective 1, Striped Bass Study (which is the same as the metadata for Objective 2, Striped Bass Study) alongside the output for the CLC Genomics Workbench Differential Expression Analyses output (i.e., statistical test results of gene expression between groups within each comparison of Grade (TG, Runts), Grade-Group (tTG, TG, R, bR), Dam (Dam A, Dam B), Sire (Sire #1-12), and Sire-Group (Large, Small). The results of these analyses are described in further detail in File no. 1.

Folders no. 6 through 10 contain the input and output files from the machine learning analyses of the RNASeq data generated from the striped bass sampled for this project. All read values (RPKM) were generated in CLC Genomics Workbench and the subsequent output that was used to perform these analyses were the “transcript expression (TE)” values. Specifically, each folder contains the following:

- CSV and ARFF versions of the WEKA input file containing all of the transcripts measured (32,018 transcripts) whereby each individual fish sampled is identified only by the group-identifier within a certain comparison (e.g., for the Dam comparison, offspring sampled are identified by “A” or “B” meaning to have been determined to be born from Dam A or Dam B based on genotyping and parentage assignment performed in year 4 of this project. These files are named: “SB-WEKA-[Comparison Name]_32018.csv” (and .arff).

- XLSX and CSV versions of only the attributes assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA for a given comparison (e.g., Dam) and the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by comparison grouping. These ranked attributes were the input for the iterative (i.e., where the top- or bottom-ranked attributes are excluded from the model) analyses of model performance (see: sub-folders). “RNASeq” in the file name is used to indicate that the format is an XLSX file, “WEKA” is used to indicate that the data contained within a given file is formatted for WEKA. These files are named: “SB-[RNASeq/WEKA]-[Comparison Name]_[number of ranked attributes]-significant.xlsx/.csv”. The InfoGainAttributeEval output file for each comparison is included in this folder as well and is named “SB-[Comparison Name]_InfoGain_Results.xlsx”.

- XLSX and CSV versions of only the attributes assigned a p-value below or equal to 0.05 via Differential Expression Analysis conducted in CLC Genomics Workbench and the associated data (RPKM transcript expression, TE, values) for each of the samples in order of 1-72, but only identified by comparison grouping. These files are named: “SB-[RNASeq/WEKA]-[Comparison Name]_[number of attributes with statistically differing expression for a given comparison]-ranked.xlsx/.csv”.

- XLSX and CSV versions of the RPKM transcript expression, TE, values for the transcripts shared between the p-value “SB-WEKA-[Comparison Name]_TopAttributes_InfoGain” containing CSV files that only include data (i.e., transcript RPKM values) from a specified number of the top-ranked of all ranked (by InfoGain in WEKA) attributes.

> “SB-WEKA-[Comparison Name]_MinusTopAttribtues_InfoGain” containing CSV files that only include data (i.e., transcript RPKM values) from a specified number of only the bottom-ranked of all ranked (by InfoGain in WEKA attributes).

> “SB-WEKA-[Comparison Name]_NegativeControl containing one XLSX file that includes additional identifiers alongside with only those transcripts and associated data (i.e., transcript RPKM values) deemed optimal after considerations of model over- and underfitting were made. These transcripts (attributes) and associated data were copied into ten CSV files in which the sample identifiers (i.e., comparison identifiers) were randomized to serve as a negative control.

Folder no. 6 contains the above sub-folders and files for the Grade comparison (Top Grade, TG, and Runts, R). Folder no. 7 contains the above for the Grade-Group comparison (top Top Grade, tTG, TG, R, and bottom Runts, bR). Folder no. 8 contains the above for the Dam comparison (Dam A, Dam B). Folder no. 9 contains the above for the Sire comparison (Sires no. 1-12). Folder no. 10 contains the above for the Sire-Group comparison (Large, Small).

File Inventory, Individual Files:

1. File Name: All-SB-RNAseq_Metadata.xlsx
Description: The metadata table with phenotypic and environmental data associated to each striped bass and subsequent sample used for the RNASeq analysis completed in CLC Genomics Workbench.
2. File Name: SB-RNASeq-ALL_32018.xlsx
Description: Transcript expression (TE) data generated in CLC Genomics Workbench for all 32,018 SB transcripts evaluated across all seventy-two (72) samples. Data is associated with RNASeq sample names and their group identifiers (Grade, Grade Group, Dam, Sire, Sire Group). Subsequent sheets contain only the RNASeq sample name and specific group identifiers. Copies were made of the group-specific sheets and used as input into R to generate transposed CSV files that could be converted into ARFF files for WEKA analysis.

5. Folder Name: SB_RNASeq-Analysis-Files

Output of analyses using the trimmed, QC RNASeq reads provided in the SB_Trimmed-QC-Paired-Reads folder (zipped file name: SB_Trimmed-Paired_Fastqfiles.zip). The RNA-Seq Analysis tool in the Transcriptomics Analysis toolbox in CLC Genomics Workbench was used to perform these analyses and this approach is based off of that taken by Mortazavi et al. (2008), full citation: Mortazavi, A., Williams, B.A., McCue, K., Schaeffer, L. and Wold, B., 2008. Mapping and quantifying mammalian transcriptomes by RNA-Seq. Nature methods, 5(7), pp.621-628.

This RNASeq analysis has been completed for the following samples 3-9, 39-52 (in CLC Workbench as: RNA Seq Analysis Jan 31_FFAR) and 53-72 (in CLC Workbench as: RNA Seq Analysis Feb 10_FFAR). The RNASeq analysis of samples 1,2,10-38 of 1-72 (in CLC Workbench as: RNA Seq Analysis Sept 18_FFAR) was completed in Year 4 of this project and the corresponding data was provided in that year (2020).
5a. Sub-Folder Name: CLC_RNASeqAnalysis_FFAR-SB-3-9_39-52.zip
File Name # Files Description
3M_S11_L002_R1_001 (FG) 1 For each sample that the RNASeq analysis has been completed (samples 3-9, 39-52 of all samples, n = 72) the following files have been generated:
- annotation track files summarizing the evidence for gene fusions, indicated by “(FG)”
- an expression track file summarizing expression at the gene level, indicated by “(GE)”
- a reads track file containing the mapping of the reads to references, indicated by “(Reads)”
- a report file providing information on the: (1) number of reads; (2) reference sequences used and their length; (3) the total number of genes and transcripts found in the reference;(4) the number of transcripts per gene (the number of mRNA annotations per gene annotation); (5) the number of exons per transcript; the distribution of transcript lengths; (6) statistics on the number of reads mapped in pairs, the number of reads in broken pairs, and the number of unmapped reads; (7) fragment counting (the total number of fragments used for calculating expression, divided into uniquely and non-specifically mapped reads), including the division of these fragments that are counted into different types; (8) the reads that map completely within an exon; (9) reads that map across an exon junction; (10) total number of reads that fall entirely within an exon or an exon-exon junction; (11) reads that fall partly or entirely within an intron; (12) all reads that map to the gene; (13) all reads that map partly or entirely between genes; (14) match specificity; and (15) graphs of the distance between mapped reads in pairs (paired reads).
- a second expression track file summarizing expression at the transcript level, indicated by “(TE)”
- two reads lists files for paired and single reads that were unmapped, indicated by “(Unmapped reads)” and “(paired)” or “(single)”, respectively
- activity logs(.csv files) for each sample. The first log is not numbered and the remaining log files are numbered 1-20 which correspond to the RNASeq analysis activity for each of the samples 3-9, 39-52 whereby the following log corresponds to the sample listed that is immediately following the arrow:

RNA-Seq Analysis log.csv → 6M_S14

RNA-Seq Analysis log-1.csv → 7M_S15

RNA-Seq Analysis log-2.csv → 5M_S13

RNA-Seq Analysis log-3.csv → 4M_S12

RNA-Seq Analysis log-4.csv → 8M_S16

RNA-Seq Analysis log-5.csv → 3M_S11

RNA-Seq Analysis log-6.csv → 39M_S47

RNA-Seq Analysis log-7.csv → 43M_S51

RNA-Seq Analysis log-8.csv → 9M_S17

RNA-Seq Analysis log-9.csv → 41M_S49

RNA-Seq Analysis log-10.csv → 40M_S48

RNA-Seq Analysis log-11.csv → 42M_S50

RNA-Seq Analysis log-12.csv → 46M_S54

RNA-Seq Analysis log-13.csv → 44M_S52

RNA-Seq Analysis log-14.csv → 45M_S53

RNA-Seq Analysis log-15.csv → 47M_S55

RNA-Seq Analysis log-16.csv → 48M_S56

RNA-Seq Analysis log-17.csv → 49M_S57

RNA-Seq Analysis log-18.csv → 51M_S59

RNA-Seq Analysis log-19.csv → 52M_S60

RNA-Seq Analysis log-20.csv → 50M_S58

*Note: the number prior to “M” is the true fish number determined at the point of sampling and the number following “S” is the sample number assigned by the NCSU GSL.

3M_S11_L002_R1_001 (GE) 1

3M_S11_L002_R1_001 (RNA-Seq reads) 1

3M_S11_L002_R1_001 (RNA-Seq report) 1

3M_S11_L002_R1_001 (TE) 1

3M_S11_L002_R1_001 (Unmapped reads)

[3M_S11_L002_R1_001] (paired) 1

3M_S11_L002_R1_001 (Unmapped reads)

```
[3M_S11_L002_R1_001] (single) 1
4M_S12_L002_R1_001 (FG) 1
4M_S12_L002_R1_001 (GE) 1
4M_S12_L002_R1_001 (RNA-Seq reads) 1
4M_S12_L002_R1_001 (RNA-Seq report) 1
4M_S12_L002_R1_001 (TE) 1
4M_S12_L002_R1_001 (Unmapped reads)
[4M_S12_L002_R1_001] (paired) 1
4M_S12_L002_R1_001 (Unmapped reads)
[4M_S12_L002_R1_001] (single) 1
5M_S13_L002_R1_001 (FG) 1
5M_S13_L002_R1_001 (GE) 1
5M_S13_L002_R1_001 (RNA-Seq reads) 1
5M_S13_L002_R1_001 (RNA-Seq report) 1
5M_S13_L002_R1_001 (TE) 1
5M_S13_L002_R1_001 (Unmapped reads)
[5M_S13_L002_R1_001] (paired) 1
5M_S13_L002_R1_001 (Unmapped reads)
[5M_S13_L002_R1_001] (single) 1
6M_S14_L002_R1_001 (FG) 1
6M_S14_L002_R1_001 (GE) 1
6M_S14_L002_R1_001 (RNA-Seq reads) 1
6M_S14_L002_R1_001 (RNA-Seq report) 1
6M_S14_L002_R1_001 (TE) 1
6M_S14_L002_R1_001 (Unmapped reads)
[6M_S14_L002_R1_001] (paired) 1
6M_S14_L002_R1_001 (Unmapped reads)
[6M_S14_L002_R1_001] (single) 1
7M_S15_L002_R1_001 (FG) 1
7M_S15_L002_R1_001 (GE) 1
7M_S15_L002_R1_001 (RNA-Seq reads) 1
7M_S15_L002_R1_001 (RNA-Seq report) 1
7M_S15_L002_R1_001 (TE) 1
7M_S15_L002_R1_001 (Unmapped reads)
[7M_S15_L002_R1_001] (paired) 1
7M_S15_L002_R1_001 (Unmapped reads)
```

```
[7M_S15_L002_R1_001] (single) 1
8M_S16_L002_R1_001 (FG) 1
8M_S16_L002_R1_001 (GE) 1
8M_S16_L002_R1_001 (RNA-Seq reads) 1
8M_S16_L002_R1_001 (RNA-Seq report) 1
8M_S16_L002_R1_001 (TE) 1
8M_S16_L002_R1_001 (Unmapped reads)
[8M_S16_L002_R1_001] (paired) 1
8M_S16_L002_R1_001 (Unmapped reads)
[8M_S16_L002_R1_001] (single) 1
9M_S17_L002_R1_001 (FG) 1
9M_S17_L002_R1_001 (GE) 1
9M_S17_L002_R1_001 (RNA-Seq reads) 1
9M_S17_L002_R1_001 (RNA-Seq report) 1
9M_S17_L002_R1_001 (TE) 1
9M_S17_L002_R1_001 (Unmapped reads)
[9M_S17_L002_R1_001] (paired) 1
9M_S17_L002_R1_001 (Unmapped reads)
[9M_S17_L002_R1_001] (single) 1
39M_S47_L002_R1_001 (FG) 1
39M_S47_L002_R1_001 (GE) 1
39M_S47_L002_R1_001 (RNA-Seq reads) 1
39M_S47_L002_R1_001 (RNA-Seq report) 1
39M_S47_L002_R1_001 (TE) 1
39M_S47_L002_R1_001 (Unmapped reads)
[39M_S47_L002_R1_001] (paired) 1
39M_S47_L002_R1_001 (Unmapped reads)
[39M_S47_L002_R1_001] (single) 1
40M_S48_L002_R1_001 (FG) 1
40M_S48_L002_R1_001 (GE) 1
40M_S48_L002_R1_001 (RNA-Seq reads) 1
40M_S48_L002_R1_001 (RNA-Seq report) 1
40M_S48_L002_R1_001 (TE) 1
40M_S48_L002_R1_001 (Unmapped reads)
[40M_S48_L002_R1_001] (paired) 1
40M_S48_L002_R1_001 (Unmapped reads)
```

[40M_S48_L002_R1_001] (single) 1

41M_S49_L002_R1_001 (FG) 1

41M_S49_L002_R1_001 (GE) 1

41M_S49_L002_R1_001 (RNA-Seq reads) 1

41M_S49_L002_R1_001 (RNA-Seq report) 1

41M_S49_L002_R1_001 (TE) 1

41M_S49_L002_R1_001 (Unmapped reads)

[41M_S49_L002_R1_001] (paired) 1

41M_S49_L002_R1_001 (Unmapped reads)

[41M_S49_L002_R1_001] (single) 1

42M_S50_L002_R1_001 (FG) 1

42M_S50_L002_R1_001 (GE) 1

42M_S50_L002_R1_001 (RNA-Seq reads) 1

42M_S50_L002_R1_001 (RNA-Seq report) 1

42M_S50_L002_R1_001 (TE) 1

42M_S50_L002_R1_001 (Unmapped reads)

[42M_S50_L002_R1_001] (paired) 1

42M_S50_L002_R1_001 (Unmapped reads)

[42M_S50_L002_R1_001] (single) 1

43M_S51_L002_R1_001 (FG) 1

43M_S51_L002_R1_001 (GE) 1

43M_S51_L002_R1_001 (RNA-Seq reads) 1

43M_S51_L002_R1_001 (RNA-Seq report) 1

43M_S51_L002_R1_001 (TE) 1

43M_S51_L002_R1_001 (Unmapped reads)

[43M_S51_L002_R1_001] (paired) 1

43M_S51_L002_R1_001 (Unmapped reads)

[43M_S51_L002_R1_001] (single) 1

44M_S52_L002_R1_001 (FG) 1

44M_S52_L002_R1_001 (GE) 1

44M_S52_L002_R1_001 (RNA-Seq reads) 1

44M_S52_L002_R1_001 (RNA-Seq report) 1

44M_S52_L002_R1_001 (TE) 1

44M_S52_L002_R1_001 (Unmapped reads)

[44M_S52_L002_R1_001] (paired) 1

44M_S52_L002_R1_001 (Unmapped reads)

[44M_S52_L002_R1_001] (single) 1

45M_S53_L002_R1_001 (FG) 1

45M_S53_L002_R1_001 (GE) 1

45M_S53_L002_R1_001 (RNA-Seq reads) 1

45M_S53_L002_R1_001 (RNA-Seq report) 1

45M_S53_L002_R1_001 (TE) 1

45M_S53_L002_R1_001 (Unmapped reads)

[45M_S53_L002_R1_001] (paired) 1

45M_S53_L002_R1_001 (Unmapped reads)

[45M_S53_L002_R1_001] (single) 1

46M_S54_L002_R1_001 (FG) 1

46M_S54_L002_R1_001 (GE) 1

46M_S54_L002_R1_001 (RNA-Seq reads) 1

46M_S54_L002_R1_001 (RNA-Seq report) 1

46M_S54_L002_R1_001 (TE) 1

46M_S54_L002_R1_001 (Unmapped reads)

[46M_S54_L002_R1_001] (paired) 1

46M_S54_L002_R1_001 (Unmapped reads)

[46M_S54_L002_R1_001] (single) 1

47M_S55_L002_R1_001 (FG) 1

47M_S55_L002_R1_001 (GE) 1

47M_S55_L002_R1_001 (RNA-Seq reads) 1

47M_S55_L002_R1_001 (RNA-Seq report) 1

47M_S55_L002_R1_001 (TE) 1

47M_S55_L002_R1_001 (Unmapped reads)

[47M_S55_L002_R1_001] (paired) 1

47M_S55_L002_R1_001 (Unmapped reads)

[47M_S55_L002_R1_001] (single) 1

48M_S56_L002_R1_001 (FG) 1

48M_S56_L002_R1_001 (GE) 1

48M_S56_L002_R1_001 (RNA-Seq reads) 1

48M_S56_L002_R1_001 (RNA-Seq report) 1

48M_S56_L002_R1_001 (TE) 1

48M_S56_L002_R1_001 (Unmapped reads)

[48M_S56_L002_R1_001] (paired) 1

48M_S56_L002_R1_001 (Unmapped reads)

[48M_S56_L002_R1_001] (single) 1

49M_S57_L002_R1_001 (FG) 1

49M_S57_L002_R1_001 (GE) 1

49M_S57_L002_R1_001 (RNA-Seq reads) 1

49M_S57_L002_R1_001 (RNA-Seq report) 1

49M_S57_L002_R1_001 (TE) 1

49M_S57_L002_R1_001 (Unmapped reads)

[49M_S57_L002_R1_001] (paired) 1

49M_S57_L002_R1_001 (Unmapped reads)

[49M_S57_L002_R1_001] (single) 1

50M_S58_L002_R1_001 (FG) 1

50M_S58_L002_R1_001 (GE) 1

50M_S58_L002_R1_001 (RNA-Seq reads) 1

50M_S58_L002_R1_001 (RNA-Seq report) 1

50M_S58_L002_R1_001 (TE) 1

50M_S58_L002_R1_001 (Unmapped reads)

[50M_S58_L002_R1_001] (paired) 1

50M_S58_L002_R1_001 (Unmapped reads)

[50M_S58_L002_R1_001] (single) 1

51M_S59_L002_R1_001 (FG) 1

51M_S59_L002_R1_001 (GE) 1

51M_S59_L002_R1_001 (RNA-Seq reads) 1

51M_S59_L002_R1_001 (RNA-Seq report) 1

51M_S59_L002_R1_001 (TE) 1

51M_S59_L002_R1_001 (Unmapped reads)

[51M_S59_L002_R1_001] (paired) 1

51M_S59_L002_R1_001 (Unmapped reads)

[51M_S59_L002_R1_001] (single) 1

52M_S60_L002_R1_001 (FG) 1

52M_S60_L002_R1_001 (GE) 1

52M_S60_L002_R1_001 (RNA-Seq reads) 1

52M_S60_L002_R1_001 (RNA-Seq report) 1

52M_S60_L002_R1_001 (TE) 1

52M_S60_L002_R1_001 (Unmapped reads)

[52M_S60_L002_R1_001] (paired) 1

52M_S60_L002_R1_001 (Unmapped reads)

[52M_S60_L002_R1_001] (single) 1

RNA-Seq Analysis log.csv 1

RNA-Seq Analysis log-1.csv 1

RNA-Seq Analysis log-2.csv 1

RNA-Seq Analysis log-3.csv 1

RNA-Seq Analysis log-5.csv 1

RNA-Seq Analysis log-6.csv 1

RNA-Seq Analysis log-7.csv 1

RNA-Seq Analysis log-8.csv 1

RNA-Seq Analysis log-9.csv 1

RNA-Seq Analysis log-10.csv 1

RNA-Seq Analysis log-11.csv 1

RNA-Seq Analysis log-12.csv 1

RNA-Seq Analysis log-13.csv 1

RNA-Seq Analysis log-14.csv 1

RNA-Seq Analysis log-15.csv 1

RNA-Seq Analysis log-16.csv 1

RNA-Seq Analysis log-17.csv 1

RNA-Seq Analysis log-18.csv 1

RNA-Seq Analysis log-19.csv 1

RNA-Seq Analysis log-20.csv 1

5b. Sub-Folder Name: CLC_RNASeqAnalysis_FFAR-SB-53-72.zip

File Name # Files Description

53M_S61_L002_R1_001 (FG) 1 For each sample that the RNASeq analysis has been completed (samples 53-72 of all samples, n = 72) the following files have been generated:

- annotation track files summarizing the evidence for gene fusions, indicated by "(FG)"
- an expression track file summarizing expression at the gene level, indicated by "(GE)"
- a reads track file containing the mapping of the reads to references, indicated by "(Reads)"
- a report file providing information on the: (1) number of reads; (2) reference sequences used and their length; (3) the total number of genes and transcripts found in the reference;(4) the number of transcripts per gene (the number of mRNA annotations per gene annotation); (5) the number of exons per transcript; the distribution of transcript lengths; (6) statistics on the number of reads mapped in pairs, the number of reads in broken pairs, and the number of unmapped reads; (7) fragment counting (the total number of fragments used for calculating expression, divided into uniquely and non-specifically mapped reads), including the division of these fragments that are counted into different types; (8) the reads that map completely within an exon; (9) reads that map across an exon junction; (10) total number

of reads that fall entirely within an exon or an exon-exon junction; (11) reads that fall partly or entirely within an intron; (12) all reads that map to the gene; (13) all reads that map partly or entirely between genes; (14) match specificity; and (15) graphs of the distance between mapped reads in pairs (paired reads).

- a second expression track file summarizing expression at the transcript level, indicated by "(TE)"
- two reads lists files for paired and single reads that were unmapped, indicated by "(Unmapped reads)" and "(paired)" or "(single)", respectively
- activity logs(.csv files) for each sample. The first log is not numbered and the remaining log files are numbered 1-19 which correspond to the RNASeq analysis activity for each of the samples 53-72 whereby the following log corresponds to the sample listed that is immediately following the arrow:

RNA-Seq Analysis log.csv → 55M_S63

RNA-Seq Analysis log-1.csv → 56M_S64

RNA-Seq Analysis log-2.csv → 53M_S61

RNA-Seq Analysis log-3.csv → 57M_S65

RNA-Seq Analysis log-4.csv → 54M_S62

RNA-Seq Analysis log-5.csv → 60M_S68

RNA-Seq Analysis log-6.csv → 61M_S69

RNA-Seq Analysis log-7.csv → 59M_S67

RNA-Seq Analysis log-8.csv → 58M_S66

RNA-Seq Analysis log-9.csv → 63M_S71

RNA-Seq Analysis log-10.csv → 62M_S70

RNA-Seq Analysis log-11.csv → 64M_S72

RNA-Seq Analysis log-12.csv → 65M_S73

RNA-Seq Analysis log-13.csv → 68M_S76

RNA-Seq Analysis log-14.csv → 67M_S75

RNA-Seq Analysis log-15.csv → 69M_S77

RNA-Seq Analysis log-16.csv → 66M_S74

RNA-Seq Analysis log-17.csv → 71M_S79

RNA-Seq Analysis log-18.csv → 70M_S78

RNA-Seq Analysis log-19.csv → 72M_S80

*Note: the number prior to "M" is the true fish number determined at the point of sampling and the number following "S" is the sample number assigned by the NCSU GSL.

53M_S61_L002_R1_001 (GE) 1

53M_S61_L002_R1_001 (RNA-Seq reads) 1

53M_S61_L002_R1_001 (RNA-Seq report) 1

53M_S61_L002_R1_001 (TE) 1

53M_S61_L002_R1_001 (Unmapped reads)

[53M_S61_L002_R1_001] (paired) 1

53M_S61_L002_R1_001 (Unmapped reads)

[53M_S61_L002_R1_001] (single) 1

54M_S62_L002_R1_001 (FG) 1

54M_S62_L002_R1_001 (GE) 1

54M_S62_L002_R1_001 (RNA-Seq reads) 1

54M_S62_L002_R1_001 (RNA-Seq report) 1

54M_S62_L002_R1_001 (TE) 1

54M_S62_L002_R1_001 (Unmapped reads)

[54M_S62_L002_R1_001] (paired) 1

54M_S62_L002_R1_001 (Unmapped reads)

[54M_S62_L002_R1_001] (single) 1

55M_S63_L002_R1_001 (FG) 1

55M_S63_L002_R1_001 (GE) 1

55M_S63_L002_R1_001 (RNA-Seq reads) 1

55M_S63_L002_R1_001 (RNA-Seq report) 1

55M_S63_L002_R1_001 (TE) 1

55M_S63_L002_R1_001 (Unmapped reads)

[55M_S63_L002_R1_001] (paired) 1

55M_S63_L002_R1_001 (Unmapped reads)

[55M_S63_L002_R1_001] (single) 1

56M_S64_L002_R1_001 (FG) 1

56M_S64_L002_R1_001 (GE) 1

56M_S64_L002_R1_001 (RNA-Seq reads) 1

56M_S64_L002_R1_001 (RNA-Seq report) 1

56M_S64_L002_R1_001 (TE) 1

56M_S64_L002_R1_001 (Unmapped reads)

[56M_S64_L002_R1_001] (paired) 1

56M_S64_L002_R1_001 (Unmapped reads)

[56M_S64_L002_R1_001] (single) 1

57M_S65_L002_R1_001 (FG) 1

57M_S65_L002_R1_001 (GE) 1

57M_S65_L002_R1_001 (RNA-Seq reads) 1

57M_S65_L002_R1_001 (RNA-Seq report) 1

57M_S65_L002_R1_001 (TE) 1

57M_S65_L002_R1_001 (Unmapped reads)

[57M_S65_L002_R1_001] (paired) 1

57M_S65_L002_R1_001 (Unmapped reads)

[57M_S65_L002_R1_001] (single) 1

58M_S66_L002_R1_001 (FG) 1

58M_S66_L002_R1_001 (GE) 1

58M_S66_L002_R1_001 (RNA-Seq reads) 1

58M_S66_L002_R1_001 (RNA-Seq report) 1

58M_S66_L002_R1_001 (TE) 1

58M_S66_L002_R1_001 (Unmapped reads)

[58M_S66_L002_R1_001] (paired) 1

58M_S66_L002_R1_001 (Unmapped reads)

[58M_S66_L002_R1_001] (single) 1

59M_S67_L002_R1_001 (FG) 1

59M_S67_L002_R1_001 (GE) 1

59M_S67_L002_R1_001 (RNA-Seq reads) 1

59M_S67_L002_R1_001 (RNA-Seq report) 1

59M_S67_L002_R1_001 (TE) 1

59M_S67_L002_R1_001 (Unmapped reads)

[59M_S67_L002_R1_001] (paired) 1

59M_S67_L002_R1_001 (Unmapped reads)

[59M_S67_L002_R1_001] (single) 1

60M_S68_L002_R1_001 (FG) 1

60M_S68_L002_R1_001 (GE) 1

60M_S68_L002_R1_001 (RNA-Seq reads) 1

60M_S68_L002_R1_001 (RNA-Seq report) 1

60M_S68_L002_R1_001 (TE) 1

60M_S68_L002_R1_001 (Unmapped reads)

[60M_S68_L002_R1_001] (paired) 1

60M_S68_L002_R1_001 (Unmapped reads)

[60M_S68_L002_R1_001] (single) 1

61M_S69_L002_R1_001 (FG) 1

61M_S69_L002_R1_001 (GE) 1

61M_S69_L002_R1_001 (RNA-Seq reads) 1

```
61M_S69_L002_R1_001 (RNA-Seq report) 1
61M_S69_L002_R1_001 (TE) 1
61M_S69_L002_R1_001 (Unmapped reads)
[61M_S69_L002_R1_001] (paired) 1
61M_S69_L002_R1_001 (Unmapped reads)
[61M_S69_L002_R1_001] (single) 1
62M_S70_L002_R1_001 (FG) 1
62M_S70_L002_R1_001 (GE) 1
62M_S70_L002_R1_001 (RNA-Seq reads) 1
62M_S70_L002_R1_001 (RNA-Seq report) 1
62M_S70_L002_R1_001 (TE) 1
62M_S70_L002_R1_001 (Unmapped reads)
[62M_S70_L002_R1_001] (paired) 1
62M_S70_L002_R1_001 (Unmapped reads)
[62M_S70_L002_R1_001] (single) 1
63M_S71_L002_R1_001 (FG) 1
63M_S71_L002_R1_001 (GE) 1
63M_S71_L002_R1_001 (RNA-Seq reads) 1
63M_S71_L002_R1_001 (RNA-Seq report) 1
63M_S71_L002_R1_001 (TE) 1
63M_S71_L002_R1_001 (Unmapped reads)
[63M_S71_L002_R1_001] (paired) 1
63M_S71_L002_R1_001 (Unmapped reads)
[63M_S71_L002_R1_001] (single) 1
64M_S72_L002_R1_001 (FG) 1
64M_S72_L002_R1_001 (GE) 1
64M_S72_L002_R1_001 (RNA-Seq reads) 1
64M_S72_L002_R1_001 (RNA-Seq report) 1
64M_S72_L002_R1_001 (TE) 1
64M_S72_L002_R1_001 (Unmapped reads) [64M_S72_L002_R1_001] (paired) 1
64M_S72_L002_R1_001 (Unmapped reads) [64M_S72_L002_R1_001] (single) 1
65M_S73_L002_R1_001 (FG) 1
65M_S73_L002_R1_001 (GE) 1
65M_S73_L002_R1_001 (RNA-Seq reads) 1
65M_S73_L002_R1_001 (RNA-Seq report) 1
65M_S73_L002_R1_001 (TE) 1
```

65M_S73_L002_R1_001 (Unmapped reads) [65M_S73_L002_R1_001] (paired) 1

65M_S73_L002_R1_001 (Unmapped reads) [65M_S73_L002_R1_001] (single) 1

66M_S74_L002_R1_001 (FG) 1

66M_S74_L002_R1_001 (GE) 1

66M_S74_L002_R1_001 (RNA-Seq reads) 1

66M_S74_L002_R1_001 (RNA-Seq report) 1

66M_S74_L002_R1_001 (TE) 1

66M_S74_L002_R1_001 (Unmapped reads)

[66M_S74_L002_R1_001] (paired) 1

66M_S74_L002_R1_001 (Unmapped reads)

[66M_S74_L002_R1_001] (single) 1

67M_S75_L002_R1_001 (FG) 1

67M_S75_L002_R1_001 (GE) 1

67M_S75_L002_R1_001 (RNA-Seq reads) 1

67M_S75_L002_R1_001 (RNA-Seq report) 1

67M_S75_L002_R1_001 (TE) 1

67M_S75_L002_R1_001 (Unmapped reads)

[67M_S75_L002_R1_001] (paired) 1

67M_S75_L002_R1_001 (Unmapped reads)

[67M_S75_L002_R1_001] (single) 1

68M_S76_L002_R1_001 (FG) 1

68M_S76_L002_R1_001 (GE) 1

68M_S76_L002_R1_001 (RNA-Seq reads) 1

68M_S76_L002_R1_001 (RNA-Seq report) 1

68M_S76_L002_R1_001 (TE) 1

68M_S76_L002_R1_001 (Unmapped reads)

[68M_S76_L002_R1_001] (paired) 1

68M_S76_L002_R1_001 (Unmapped reads)

[68M_S76_L002_R1_001] (single) 1

69M_S77_L002_R1_001 (FG) 1

69M_S77_L002_R1_001 (GE) 1

69M_S77_L002_R1_001 (RNA-Seq reads) 1

69M_S77_L002_R1_001 (RNA-Seq report) 1

69M_S77_L002_R1_001 (TE) 1

69M_S77_L002_R1_001 (Unmapped reads)

[69M_S77_L002_R1_001] (paired) 1

69M_S77_L002_R1_001 (Unmapped reads)

[69M_S77_L002_R1_001] (single) 1

70M_S78_L002_R1_001 (FG) 1

70M_S78_L002_R1_001 (GE) 1

70M_S78_L002_R1_001 (RNA-Seq reads) 1

70M_S78_L002_R1_001 (RNA-Seq report) 1

70M_S78_L002_R1_001 (TE) 1

70M_S78_L002_R1_001 (Unmapped reads)

[70M_S78_L002_R1_001] (paired) 1

70M_S78_L002_R1_001 (Unmapped reads)

[70M_S78_L002_R1_001] (single) 1

71M_S79_L002_R1_001 (FG) 1

71M_S79_L002_R1_001 (GE) 1

71M_S79_L002_R1_001 (RNA-Seq reads) 1

71M_S79_L002_R1_001 (RNA-Seq report) 1

71M_S79_L002_R1_001 (TE) 1

71M_S79_L002_R1_001 (Unmapped reads)

[71M_S79_L002_R1_001] (paired) 1

71M_S79_L002_R1_001 (Unmapped reads)

[71M_S79_L002_R1_001] (single) 1

72M_S80_L002_R1_001 (FG) 1

72M_S80_L002_R1_001 (GE) 1

72M_S80_L002_R1_001 (RNA-Seq reads) 1

72M_S80_L002_R1_001 (RNA-Seq report) 1

72M_S80_L002_R1_001 (TE) 1

72M_S80_L002_R1_001 (Unmapped reads)

[72M_S80_L002_R1_001] (paired) 1

72M_S80_L002_R1_001 (Unmapped reads)

[72M_S80_L002_R1_001] (single) 1

RNA-Seq Analysis log.csv 1

RNA-Seq Analysis log-1.csv 1

RNA-Seq Analysis log-2.csv 1

RNA-Seq Analysis log-3.csv 1

RNA-Seq Analysis log-5.csv 1

RNA-Seq Analysis log-6.csv 1

RNA-Seq Analysis log-7.csv 1

RNA-Seq Analysis log-8.csv 1

RNA-Seq Analysis log-9.csv 1

RNA-Seq Analysis log-10.csv 1

RNA-Seq Analysis log-11.csv 1

RNA-Seq Analysis log-12.csv 1

RNA-Seq Analysis log-13.csv 1

RNA-Seq Analysis log-14.csv 1

RNA-Seq Analysis log-15.csv 1

RNA-Seq Analysis log-16.csv 1

RNA-Seq Analysis log-17.csv 1

RNA-Seq Analysis log-18.csv 1

RNA-Seq Analysis log-19.csv 1

6. Folder Name: SB_RNASeq-CLC-Statistical-Analyses_Output

File Name # Files Description

All-SB-RNASeq_Metadata.xlsx 1 The metadata associated with each individual SB, identified by the RNASeq file name. This file was input into CLC Genomics Workbench to conduct statistical analyses and used throughout the project. The sheet "SB Metadata" includes these identifiers and information such as fish weight and tank. The sheet "SB Metadata - TE values" contains the transcript expression (TE) values generated in CLC Genomics workbench for each of the 32,018 identified transcripts with all associated group identifiers (e.g., Sire).

SB-RNASeq_CLC-Stats_GRADE.xlsx 1 The results output from the statistical tests completed in CLC Genomics Workbench ("Differential Expression for RNA-Seq") using the SB RNASeq data generated for this project (provided in year 4 and here in year 5) and testing for differences between Grade ("Top Grade" and "Runts"). Other sheets in the workbook file include the isolated lists of (1) transcripts determined to have a p value less than or equal to 0.05, (2) a p value greater than 0.05 and less than 0.10, and (3) a p value above 0.10.

SB-RNASeq_CLC-Stats_GRADE-GROUP.xlsx 1 The results output from the statistical tests completed in CLC Genomics Workbench ("Differential Expression for RNA-Seq") using the SB RNASeq data generated for this project (provided in year 4 and here in year 5) and testing for differences between Grade Group ("top Top Grade", "Top Grade", "Runts", "bottom Runts"). Other sheets in the workbook file include the isolated lists of (1) transcripts determined to have a p value less than or equal to 0.05, (2) a p value greater than 0.05 and less than 0.10, and (3) a p value above 0.10. The other sheets in the workbook include the results of the pairwise comparisons made between each group (e.g., Top Grade versus bottom Runts, Top Grade versus Runts, and so on).

SB-RNASeq_CLC-Stats_DAM.xlsx 1 The results output from the statistical tests completed in CLC Genomics Workbench ("Differential Expression for RNA-Seq") using the SB RNASeq data generated for this project (provided in year 4 and here in year 5) and testing for differences between Dam (“Dam A” and “Dam B”). Other sheets in the workbook file include the isolated lists of (1) transcripts determined to have a p value less than or equal to 0.05, (2) a p value greater than 0.05 and less than 0.10, and (3) a p value above 0.10.

SB-RNASeq_CLC-Stats_SIRE.xlsx 1 The results output from the statistical tests completed in CLC Genomics Workbench (“Differential Expression for RNA-Seq”) using the SB RNASeq data generated for this project (provided in year 4 and here in year 5) and testing for differences between Sire (“SB-S_1-L”, “SB-S_2-L”, “SB-S_3-L”, “SB-S_7-L”, “SB-S_8-L”, “SB_S_9_L”, “SB-S_4-S”, “SB-S_5-S”, “SB-S_6-S”, “SB-S_10-S”, “SB-S_11-S”, and “SB-S_12-S”). Other sheets in the workbook file include the isolated lists of (1) transcripts determined to have a p value less than or equal to 0.05, (2) a p value greater than 0.05 and less than 0.10, and (3) a p value above 0.10.

SB-RNASeq_CLC-Stats_SIRE-GROUP.xlsx 1 The results output from the statistical tests completed in CLC Genomics Workbench (“Differential Expression for RNA-Seq”) using the SB RNASeq data generated for this project (provided in year 4 and here in year 5) and testing for differences between Sire Group (“Small” and “Large”). Other sheets in the workbook file include the isolated lists of (1) transcripts determined to have a p value less than or equal to 0.05, (2) a p value greater than 0.05 and less than 0.10, and (3) a p value above 0.10.

6a. Sub-Folder Name: Sire Group Pairwise Comparisons Output

File Name # Files Description

[Sire Name 1-12-L/S] vs. [Sire Name 1-12-L/S].xlsx 66 The specific results of each pairwise comparison conducted using CLC Genomics Workbench between individual sires (twelve sires total). Each file has the name of the two sires for which the results are represented. The second sheet in each workbook, “p < 0.05” contains information on only the transcripts that had a p value equal to or less than 0.05.

7. Folder Name: SB-WEKA-Analysis_GRADE

File Name # Files Description

SB-WEKA_Output-GRADE.docx 1 A .DOCX file containing the copied-and-pasted summary output from each algorithm run in WEKA using data associated with GRADE (TG, R) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists).

SB-WEKA_Results-GRADE.xlsx 1 A .XLSX file containing the copied-and-pasted percent correct classification, kappa statistic, and AUROC output from each algorithm run in WEKA using data associated with GRADE (TG, R) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists). InfoGain AttributeEval results and figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-WEKA-GRADE_Results-SUMMARY.docx 1 A .DOCX file summarizing the percent correct classification output from each algorithm run in WEKA using data associated with GRADE (TG, R) identifiers (attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset). Figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-RNASeq-GRADE_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and GRADE identifiers (TG, Runts).

SB-WEKA-GRADE_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and GRADE identifiers (TG, Runts) in WEKA-input format (identifiers are in the columns and transcript IDs are row-headers). This file was transposed in R Studio, as the number of columns necessary exceeds the capacity of Microsoft Excel.

SB-WEKA-GRADE_32018.arff

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and GRADE identifiers (TG, Runts) in WEKA format (ARFF files). (identifiers are in the columns and transcript IDs are row-headers).

SB-GRADE_InfoGain_Results.xlsx The results of the Information Gain value ("score") assignment of all 32,018 attributes for the Grade comparison performed in WEKA. Briefly, WEKA evaluates the information gained (Shannon's Entropy) of each attribute for a given comparison, Grade via the WEKA tool InfoGainAttributeEval (SVM → SMO algorithm).

SB-RNASeq-GRADE_965-ranked.xlsx

1 An excel file containing only the attributes assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 965 attributes) and the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by Grade grouping (TG, Runt). The rank order (1-965) for each attribute is included as the top row and was used to generate the files that only include a certain number of the top ranked attributes (see: sub-folders).

SB-RNASeq-GRADE_3193-significant.xlsx

1 An excel file containing only the attributes assigned a p-value below or equal to 0.05 (n = 3,193 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench (see: "SB-RNASeq_CLC-Stats_GRADE.xlsx" in the Folder: "SB_RNASeq-CLC-Statistical-Analyses_Output") and the associated data (RPKM transcript expression, TE, values) for each of the samples in order of 1-72, but

only identified by Grade grouping (TG, Runt). The rank order (1-3,193) for each attribute is included.

SB-WEKA-GRADE_3193-significant.csv

1 A WEKA-formatted CSV file containing only the attributes assigned a p-value below or equal to 0.05 (n = 3,193 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench, the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by Grade grouping (TG, Runt).

SB-RNASeq_GRADE_3723-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 3,723 unique transcripts (435 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 965 attributes), (2) assigned a p-value below or equal to 0.05 (n = 3,193 attributes), or (3) both. Samples (1-72) are identified by Grade grouping (TG, Runt). Transcripts are in alphabetical order.

SB-WEKA_GRADE_3723-ranked-signif.csv 1 A WEKA-formatted CSV file containing the 3,723 unique attributes (435 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 965 attributes), (2) assigned a p-value below or equal to 0.05 (n = 3,193 attributes), or (3) both. Samples (1-72) are identified by Grade grouping (TG, Runt). Transcripts are in alphabetical order.

SB-RNASeq_GRADE_435-shared-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 435 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by Grade grouping (TG, Runt). Transcripts are in alphabetical order.

SB-WEKA_GRADE_435-shared-ranked-signif.csv 1 A WEKA formatted CSV file containing the RPKM transcript expression, TE, values for the 435 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by Grade grouping (TG, Runt). Transcripts are in alphabetical order.

SB-GRADE_InfoGain-StatSignif_Results.xlsx 1 An excel file containing the following information on each sheet:

[p <|= 0.05] - the CLC Genomics Workbench descriptors of the transcripts that had a p-value less than or equal to 0.05 in the differential expression analysis between Grades (TG, Runt) and including p-value order (1-3193 from smallest to largest).

[Ranked InfoGain] - the WEKA output for the 965 transcripts assigned a InfoGain (Shannon's Entropy) value above 0.0 for the Grade comparison and including the rank order (1-965 from most to least informative among this set)

[Shared - InfoGain and p value] - CLC Genomics Workbench descriptors of the 435 transcripts that were assigned a p-value less than or equal to 0.05 and an InfoGain rank above 0.0 and including InfoGain rank and p-value orders.

[Unique - InfoGain] - CLC Genomics Workbench descriptors of the 530 transcripts that were assigned an

InfoGain rank above 0.0, but that were not assigned a p-value less than or equal to 0.05 and including InfoGain rank orders.

[Unique - pvalue] - CLC Genomics Workbench descriptors of the 2758 transcripts that were assigned a p-value less than or equal to 0.05, but that were not assigned an InfoGain rank above 0.0 and including p-value orders.

GRADE-FullList-Annotation-Evalues.xlsx 1 An excel file containing the following information about the 3,723 unique transcripts from the combined p-value and InfoGain-ranked lists for the GRADE (TG, Runt) comparison (4,158 transcripts total, 435 of which are on both lists, or are “shared”):

- Sample IDs for Individual SB 1-72
- GRADE groupings (TG, Runt)
- CLC Genomics Workbench Differential Expression Analysis output (Chromosome, Region, Max group mean, Log2 fold change, fold change, p-value, FDR p-value, Bonferoni, p-value order),
- InfoGain AttributeEval output (InfoGain Rank Order, InfoGain Value, Original Row from input)

Transcripts were assigned to a list group of “GRADE-InfoGain_p-value” (highlighted in green) if shared between the p-value and InfoGain list (n = 435 of 3,723), “GRADE-p-value” (pink) if unique (i.e., not shared with InfoGain list) to p-value </= 0.05 list (n = 2,758 of 3,723); or “GRADE-InfoGain” (blue) if unique to InfoGain list (n = 530 of 3,723).

7a.Sub-Folder Name: SB-WEKA_GRADE_TopAttributes-InfoGain

File Name # Files Description

SB-WEKA-GRADE_Top-1.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the GRADE comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 965 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA whereby bottom-ranked (of the 965) attributes were removed iteratively from the input data from the inclusion of all 965 to only the top-ranked attribute.

SB-WEKA-GRADE_Top-2.csv 1

SB-WEKA-GRADE_Top-3.csv 1

SB-WEKA-GRADE_Top-4.csv 1

SB-WEKA-GRADE_Top-5.csv 1

SB-WEKA-GRADE_Top-6.csv 1

SB-WEKA-GRADE_Top-7.csv 1

SB-WEKA-GRADE_Top-8.csv 1

SB-WEKA-GRADE_Top-9.csv 1

SB-WEKA-GRADE_Top-10.csv 1

SB-WEKA-GRADE_Top-15.csv 1

SB-WEKA-GRADE_Top-20.csv 1

SB-WEKA-GRADE_Top-25.csv 1

SB-WEKA-GRADE_Top-50.csv 1

SB-WEKA-GRADE_Top-75.csv 1

SB-WEKA-GRADE_Top-100.csv 1

SB-WEKA-GRADE_Top-150.csv 1

SB-WEKA-GRADE_Top-200.csv 1

SB-WEKA-GRADE_Top-250.csv 1

SB-WEKA-GRADE_Top-300.csv 1

SB-WEKA-GRADE_Top-350.csv 1

SB-WEKA-GRADE_Top-400.csv 1

SB-WEKA-GRADE_Top-450.csv 1

SB-WEKA-GRADE_Top-500.csv 1

SB-WEKA-GRADE_Top-550.csv 1

SB-WEKA-GRADE_Top-600.csv 1

SB-WEKA-GRADE_Top-650.csv 1

SB-WEKA-GRADE_Top-700.csv 1

SB-WEKA-GRADE_Top-750.csv 1

SB-WEKA-GRADE_Top-800.csv 1

SB-WEKA-GRADE_Top-850.csv 1

SB-WEKA-GRADE_Top-900.csv 1

SB-WEKA-GRADE_965-ranked.csv 1

7b. Sub-Folder Name: SB-WEKA_GRADE_MinusTopAttributes-InfoGain

File Name # Files Description

SB-WEKA-GRADE_Minus-Top-1.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the GRADE comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 965 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA whereby top-ranked (of the 965) attributes were removed iteratively from the input data from the inclusion of all 965 to only the bottom-ranked attributes. For example, in SB-WEKA-GRADE_Minus-Top-2.csv, the data for 963 of the ranked attributes numbers 3-965 are included, but the top number 1 and 2 ranked attributes are excluded.

SB-WEKA-GRADE_Minus-Top-2.csv 1

SB-WEKA-GRADE_Minus-Top-3.csv 1

SB-WEKA-GRADE_Minus-Top-4.csv 1

SB-WEKA-GRADE_Minus-Top-5.csv 1

SB-WEKA-GRADE_Minus-Top-6.csv 1

SB-WEKA-GRADE_Minus-Top-7.csv 1 Description

SB-WEKA-GRADE_Minus-Top-8.csv 1

SB-WEKA-GRADE_Minus-Top-9.csv 1

SB-WEKA-GRADE_Minus-Top-10.csv 1 Description

SB-WEKA-GRADE_Minus-Top-15.csv 1

SB-WEKA-GRADE_Minus-Top-20.csv 1

SB-WEKA-GRADE_Minus-Top-25.csv 1

SB-WEKA-GRADE_Minus-Top-50.csv 1 Description

SB-WEKA-GRADE_Minus-Top-100.csv 1 The cover letter and description of fin clip samples sent to the Center for Aquaculture Technologies (CAT; San Diego, CA, USA) for microsatellite genotyping of the striped bass parents and offspring sampled for this FFAR project.

SB-WEKA-GRADE_Minus-Top-150.csv 1 The sample manifest of fin clip samples sent to the Center for Aquaculture Technologies (CAT; San Diego, CA, USA) for microsatellite genotyping of the striped bass parents and offspring sampled for this FFAR project.

SB-WEKA-GRADE_Minus-Top-200.csv 1 The summary report provided by the Center for Aquaculture Technologies (CAT; San Diego, CA, USA) after completing microsatellite genotyping of the striped bass parents and offspring sampled for this FFAR project.

SB-WEKA-GRADE_Minus-Top-250.csv 1 The raw microsatellite genotyping data provided by the Center for Aquaculture Technologies (CAT; San Diego, CA, USA) and generated from the striped bass parents and offspring sampled for this FFAR project.

SB-WEKA-GRADE_Minus-Top-300.csv 1 An excel file providing information on the determined parental assignment of each striped bass offspring (1-72) based on microsatellite genotyping data analyzed in Cervus. Within this workbook the sheet “Maternity” designates which of the two potential dams produced each individual offspring; the sheet “Dam 1-A_Paternity” designates which of the six potential sires spawned with Dam 1-A (Sires 1-L, 2-L, 3-L, 4-S, 5-S, and 6-S) where the determined sires of the offspring established to have been produced by Dam 1-A; the sheet “Dam2-B_Paternity” designates which of the six potential sires spawned with Dam 2-B (Sires 7-L, 8-L, 9-L, 10-S, 11-S, and 12-S) where the determined sires of the offspring established to have been produced by Dam 2-B; the sheets “Sorted by Dam” and “Sorted by Sire” provide the parentage data alongside metadata (i.e., phenotype measured by length and weight, tank, grade) as sorted by Dam and Sire, respectively; the sheet “Notes” includes the percentage breakdown of offspring produced by each dam and sire and further percentage breakdown of offspring produced from each grade.

SB-WEKA-GRADE_Minus-Top-350.csv 1 A word document further describing the percentage breakdowns of production of each dam and sire regarding offspring striped bass grouped by grade and the statistical tests and resulting findings of comparisons conducted to determine the relationships between parental size (length, weight) and offspring size (length, weight).

SB-WEKA-GRADE_Minus-Top-400.csv 1 Description

SB-WEKA-GRADE_Minus-Top-450.csv 1

SB-WEKA-GRADE_Minus-Top-500.csv 1

SB-WEKA-GRADE_Minus-Top-550.csv 1 Description

SB-WEKA-GRADE_Minus-Top-600.csv 1

SB-WEKA-GRADE_Minus-Top-650.csv 1

SB-WEKA-GRADE_Minus-Top-700.csv 1

SB-WEKA-GRADE_Minus-Top-750.csv 1

SB-WEKA-GRADE_Minus-Top-800.csv 1

SB-WEKA-GRADE_Minus-Top-850.csv 1

SB-WEKA-GRADE_Minus-Top-900.csv 1

7c. Sub-Folder Name: SB-WEKA_GRADE_NegativeControl

File Name # Files Description

SB-RNASeq-GRADE_300-Optimal-NegCtrl.xlsx 1 The template .XLSX file for the GRADE comparison (TG, Runts) negative control. The optimal dataset (determined based on over- and underfitting data reduction) of the top 300 attributes is included on the sheet “Optimal 300” and these values are on the sheet entitled “Transposed”. There are 36 samples designated as TG and 36 samples designated as Runts in the study.

SB-WEKA-GRADE_NegCtrl-1.csv 1 A .CSV file containing negative control run number 1 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-2.csv 1 A .CSV file containing negative control run number 2 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-3.csv 1 A .CSV file containing negative control run number 3 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-4.csv 1 A .CSV file containing negative control run number 4 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-5.csv 1 A .CSV file containing negative control run number 5 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-6.csv 1 A .CSV file containing negative control run number 6 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-7.csv 1 A .CSV file containing negative control run number 7 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-8.csv 1 A .CSV file containing negative control run number 8 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-9.csv 1 A .CSV file containing negative control run number 9 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE_NegCtrl-10.csv 1 A .CSV file containing negative control run number 10 of 10. The RAND() function was used in Excel to randomize the sample designations.

8. Folder Name: SB-WEKA-Analysis_GRADE-GROUP

File Name # Files Description

SB-WEKA_Output-GRADE-GROUP.docx 1 A .DOCX file containing the copied-and-pasted summary output from each algorithm run in WEKA using data associated with GRADE GROUP (tTG, TG, R, bR) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists).

SB-WEKA_Results-GRADE-GROUP.xlsx 1 A .XLSX file containing the copied-and-pasted percent correct classification, kappa statistic, and AUROC output from each algorithm run in WEKA using data associated with GRADE GROUP (tTG, TG, R, bR) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists). InfoGain AttributeEval results and figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-WEKA-GRADE-GROUP_Results-SUMMARY.docx 1 A .DOCX file summarizing the percent correct classification output from each algorithm run in WEKA using data associated with GRADE GROUP (tTG, TG, R, bR) identifiers (attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset). Figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-RNASeq-GRADE-GROUP_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and GRADE-GROUP identifiers (tTG, TG, R, bR).

SB-WEKA-GRADEGROUP_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and GRADE GROUP identifiers (tTG, TG, R, bR) in WEKA-input format (identifiers are in the columns and transcript IDs are row-headers). This file was transposed in R

Studio, as the number of columns necessary exceeds the capacity of Microsoft Excel.

SB-WEKA-GRADEGROUP_32018.arff

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and GRADE GROUP identifiers (tTG, TG, R, bR) in WEKA format (ARFF files). (identifiers are in the columns and transcript IDs are row-headers).

SB-GRADE-GROUP_InfoGain_Results.xlsx 1 The results of the Information Gain value (“score”) assignment of all 32,018 attributes for the GRADE GROUP comparison performed in WEKA. Briefly, WEKA evaluates the information gained (Shannon’s Entropy) of each attribute for a given comparison, GRADE GROUP via the WEKA tool InfoGainAttributeEval (SVM → SMO algorithm).

SB-RNASeq-GRADE-GROUP_284-ranked.xlsx

1 An excel file containing only the attributes assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 284 attributes) and the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by Grade Group assignment (tTG, TG, Runt, bR). The rank order (1-284) for each attribute is included as the top row and was used to generate the files that only include a certain number of the top ranked attributes (see: sub-folders).

SB-RNASeq-GRADE-GROUP_621-significant.xlsx

1 An excel file containing only the attributes assigned a p-value below or equal to 0.05 (n = 621 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench (see: “SB-RNASeq_CLC-Stats_GRADE-GROUP.xlsx” in the Folder: “SB_RNASeq-CLC-Statistical-Analyses_Output”) and the associated data (RPKM transcript expression, TE, values) for each of the samples in order of 1-72, but only identified by Grade Group grouping (tTG, TG, Runt, bR). The rank order (1-621) for each attribute is included.

SB-WEKA-GRADE-GROUP_621-significant.csv

1 A WEKA-formatted CSV file containing only the attributes assigned a p-value below or equal to 0.05 (n = 621 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench, the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by Grade Group grouping (tTG, TG, Runt, bR).

SB-RNASeq_GRADE-GROUP_862-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 862 unique transcripts (43 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 284 attributes), (2) assigned a p-value below or equal to 0.05 (n = 621 attributes), or (3) both. Samples (1-72) are identified by Grade Group grouping (tTG, TG, Runt, bR). Transcripts are in alphabetical order.

SB-WEKA_GRADE-GROUP_862-ranked-signif.csv 1 A WEKA-formatted CSV file containing the 862 unique attributes (43 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 284 attributes), (2) assigned a p-value below or equal to 0.05 (n = 621 attributes), or (3) both. Samples (1-72) are identified by Grade Group grouping (tTG, TG, Runt, bR). Transcripts are in alphabetical order.

SB-RNASeq_GRADE-GROUP_43-shared-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 43 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by Grade Group grouping (tTG, TG, Runt, bR). Transcripts are in alphabetical order.

SB-WEKA_GRADE-GROUP_43-shared-ranked-signif.csv 1 A WEKA formatted CSV file containing the RPKM transcript expression, TE, values for the 43 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) identified by Grade Group grouping (tTG, TG, Runt, bR). Transcripts are in alphabetical order.

SB-GRADE-GROUP_InfoGain-StatSignif_Results.xlsx 1 An excel file containing the following information on each sheet:

[p <|= 0.05] - the CLC Genomics Workbench descriptors of the transcripts that had a p-value less than or equal to 0.05 in the differential expression analysis between Grade Groups (tTG, TG, Runt, BR) and including p-value order (1-621 from smallest to largest).

[Ranked InfoGain] - the WEKA output for the 284 transcripts assigned a InfoGain (Shannon’s Entropy) value above 0.0 for the Grade Group comparison and including the rank order (1-284 from most to least informative among this set)

[Shared - InfoGain and p value] - CLC Genomics Workbench descriptors of the 43 transcripts that were assigned a p-value less than or equal to 0.05 and an InfoGain rank above 0.0 and including InfoGain rank and p-value orders.

[Unique - InfoGain] - CLC Genomics Workbench descriptors of the 241 transcripts that were assigned an InfoGain rank above 0.0, but that were not assigned a p-value less than or equal to 0.05 and including InfoGain rank orders.

[Unique - pvalue] - CLC Genomics Workbench descriptors of the 578 transcripts that were assigned a p-value less than or equal to 0.05, but that were not assigned an InfoGain rank above 0.0 and including p-value orders.

GRADE-GROUP-FullList-Annotation-Evalues.xlsx 1 An excel file containing the following information about the 862 unique transcripts from the combined p-value and InfoGain-ranked lists for the GRADE GROUP (tTG, TG, Runt, bR) comparison (905 transcripts total, 43 of which are on both lists, or are “shared”):

- Sample IDs for Individual SB 1-72
- GRADE GROUP groupings (tTG, TG, Runt, bR)
- CLC Genomics Workbench Differential Expression Analysis output (Chromosome, Region, Max group

mean, Log2 fold change, fold change, p-value, FDR p-value, Bonferoni, p-value order),

- InfoGain AttributeEval output (InfoGain Rank Order, InfoGain Value, Original Row from input)

Transcripts were assigned to a list group of “GRADE-GROUP-InfoGain_p-value” (green) if shared between the p-value and InfoGain list (n = 43 of 862), “GRADE-GROUP-p-value” (pink) if unique (i.e., not shared with InfoGain list) to p-value </= 0.05 list (n = 578 of 862); or “GRADE-GROUP-InfoGain” (blue) if unique to InfoGain list (n = 241 of 862).

8a.Sub-Folder Name: SB-WEKA_GRADE-GROUP_TopAttributes-InfoGain

File Name # Files Description

SB-WEKA-GRADE-GROUP_284-ranked.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the GRADE GROUP comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 284 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA whereby bottom-ranked (of the 284) attributes were removed iteratively from the input data from the inclusion of all 284 to only the top-ranked attribute.

SB-WEKA-GRADE-GROUP_Top-1.csv 1

SB-WEKA-GRADE-GROUP_Top-2.csv 1

SB-WEKA-GRADE-GROUP_Top-3.csv 1

SB-WEKA-GRADE-GROUP_Top-4.csv 1

SB-WEKA-GRADE-GROUP_Top-5.csv 1

SB-WEKA-GRADE-GROUP_Top-6.csv 1

SB-WEKA-GRADE-GROUP_Top-7.csv 1

SB-WEKA-GRADE-GROUP_Top-8.csv 1

SB-WEKA-GRADE-GROUP_Top-9.csv 1

SB-WEKA-GRADE-GROUP_Top-10.csv 1

SB-WEKA-GRADE-GROUP_Top-15.csv 1

SB-WEKA-GRADE-GROUP_Top-20.csv 1

SB-WEKA-GRADE-GROUP_Top-25.csv 1

SB-WEKA-GRADE-GROUP_Top-50.csv 1

SB-WEKA-GRADE-GROUP_Top-75.csv 1

SB-WEKA-GRADE-GROUP_Top-100.csv 1

SB-WEKA-GRADE-GROUP_Top-150.csv 1

SB-WEKA-GRADE-GROUP_Top-200.csv 1

SB-WEKA-GRADE-GROUP_Top-250.csv 1

8b. Sub-Folder Name: SB-WEKA_GRADE-GROUP_MinusTopAttributes-InfoGain

File Name # Files Description

SB-WEKA-GRADE-GROUP_Minus-Top-1.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the GRADE GROUP comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 284 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA whereby top-ranked (of the 284) attributes were removed iteratively from the input data from the inclusion of all 284 to only the bottom-ranked attributes.

SB-WEKA-GRADE-GROUP_Minus-Top-2.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-3.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-4.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-5.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-6.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-7.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-8.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-9.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-10.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-15.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-20.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-25.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-50.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-75.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-100.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-150.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-200.csv 1

SB-WEKA-GRADE-GROUP_Minus-Top-250.csv 1

8c. Sub-Folder Name: SB-WEKA_GRADE-GROUP_NegativeControl

File Name # Files Description

SB-RNASeq-GRADE-GROUP_284-Optimal-NegCtrl.xlsx 1 The template .XLSX file for the GRADE-GROUP comparison (tTG, TG, R, bR) negative control. The optimal dataset (determined based on over- and underfitting data reduction) of the top 284 attributes is included on the sheet “Optimal 284” and these values are on the sheet entitled “Transposed”. There are 12 samples designated as top-Top Grade (tTG), 24 as Top Grade (TG), 24 as Runts (R), and 12 as bottom-Runts (bR) in the study.

SB-WEKA-GRADE-GROUP_NegCtrl-1.csv 1 A .CSV file containing negative control run number 1 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-2.csv 1 A .CSV file containing negative control run number 2 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-3.csv 1 A .CSV file containing negative control run number 3 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-4.csv 1 A .CSV file containing negative control run number 4 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-5.csv 1 A .CSV file containing negative control run number 5 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-6.csv 1 A .CSV file containing negative control run number 6 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-7.csv 1 A .CSV file containing negative control run number 7 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-8.csv 1 A .CSV file containing negative control run number 8 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-9.csv 1 A .CSV file containing negative control run number 9 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-GRADE-GROUP_NegCtrl-10.csv 1 A .CSV file containing negative control run number 10 of 10. The RAND() function was used in Excel to randomize the sample designations.

9. Folder Name: SB-WEKA-Analysis_DAM

File Name # Files Description

SB-WEKA_Output-DAM.docx 1 A .DOCX file containing the copied-and-pasted summary output from each algorithm run in WEKA using data associated with DAM (A, B) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists).

SB-WEKA_Results-DAM.xlsx 1 A .XLSX file containing the copied-and-pasted percent correct classification, kappa statistic, and AUROC output from each algorithm run in WEKA using data associated with DAM (A, B) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists). InfoGain AttributeEval results and figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-WEKA-DAM_Results-SUMMARY.docx 1 A .DOCX file summarizing the percent correct classification output from each algorithm run in WEKA using data associated with DAM (A, B) identifiers (attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset). Figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-RNASeq-DAM_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and DAM identifiers (A, B).

SB-WEKA-DAM_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and DAM identifiers (A, B) in WEKA-input format (identifiers are in the columns and transcript IDs are row-headers). This file was transposed in R Studio, as the number of columns necessary exceeds the capacity of Microsoft Excel.

SB-WEKA-DAM_32018.arff

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and DAM identifiers (A, B) in WEKA format (ARFF files). (identifiers are in the columns and transcript IDs are row-headers).

SB-DAM_InfoGain_Results.xlsx The results of the Information Gain value (“score”) assignment of all 32,018 attributes for the DAM comparison performed in WEKA. Briefly, WEKA evaluates the information gained (Shannon’s Entropy) of each attribute for a given comparison, DAM via the WEKA tool InfoGainAttributeEval (SVM → SMO algorithm).

SB-RNASeq-DAM_257-ranked.xlsx

1 An excel file containing only the attributes assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 257 attributes) and the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by DAM grouping (A, B). The rank order (1-257) for each attribute is included as the top row and was used to generate the files that only include a certain number of the top ranked attributes (see: sub-folders).

SB-RNASeq-DAM_1643-significant.xlsx

1 An excel file containing only the attributes assigned a p-value below or equal to 0.05 (n = 1,643 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench (see: “SB-RNASeq_CLC-Stats_DAM.xlsx” in the Folder: “SB_RNASeq-CLC-Statistical-Analyses_Output”) and the

associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by DAM grouping (A, B). The rank order (1-1,643) for each attribute is included.

SB-WEKA-DAM_1643-significant.csv

1 A WEKA-formatted CSV file containing only the attributes assigned a p-value below or equal to 0.05 (n = 1,643 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench, the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by DAM grouping (A, B).

SB-RNASeq_DAM_1751-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 1,751 unique transcripts (149 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 257 attributes), (2) assigned a p-value below or equal to 0.05 (n = 1,643 attributes), or (3) both. Samples (1-72) are identified by DAM grouping (A, B). Transcripts are in alphabetical order.

SB-WEKA_DAM_1751-ranked-signif.csv 1 A WEKA-formatted CSV file containing the 1,751 unique attributes (149 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 149 attributes), (2) assigned a p-value below or equal to 0.05 (n = 1,643 attributes), or (3) both. Samples (1-72) are identified by DAM grouping (A, B). Transcripts are in alphabetical order.

SB-RNASeq_DAM_149-shared-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 149 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by DAM grouping (A, B). Transcripts are in alphabetical order.

SB-WEKA_DAM_149-shared-ranked-signif.csv 1 A WEKA formatted CSV file containing the RPKM transcript expression, TE, values for the 149 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by DAM grouping (A, B). Transcripts are in alphabetical order.

SB-DAM_InfoGain-StatSignif_Results.xlsx 1 An excel file containing the following information on each sheet:

[p <|= 0.05] - the CLC Genomics Workbench descriptors of the transcripts that had a p-value less than or equal to 0.05 in the differential expression analysis between Dams (A, B) and including p-value order (1-1643 from smallest to largest).

[Ranked InfoGain] - the WEKA output for the 257 transcripts assigned a InfoGain (Shannon’s Entropy) value above 0.0 for the DAM comparison and including the rank order (1-257 from most to least informative among this set)

[Shared - InfoGain and p value] - CLC Genomics Workbench descriptors of the 149 transcripts that were assigned a p-value less than or equal to 0.05 and an InfoGain rank above 0.0 and including InfoGain rank and p-value orders.

[Unique - InfoGain] - CLC Genomics Workbench descriptors of the 108 transcripts that were assigned an

InfoGain rank above 0.0, but that were not assigned a p-value less than or equal to 0.05 and including InfoGain rank orders.

[Unique - pvalue] - CLC Genomics Workbench descriptors of the 1494 transcripts that were assigned a p-value less than or equal to 0.05, but that were not assigned an InfoGain rank above 0.0 and including p-value orders.

DAM-FullList-Annotation-Evalues.xlsx 1 An excel file containing the following information about the 1751 unique transcripts from the combined p-value and InfoGain-ranked lists for the DAM (A, B) comparison (1,900 transcripts total, 149 of which are on both lists, or are “shared”):

- Sample IDs for Individual SB 1-72
- DAM groupings (A, B)
- CLC Genomics Workbench Differential Expression Analysis output (Chromosome, Region, Max group mean, Log2 fold change, fold change, p-value, FDR p-value, Bonferoni, p-value order),
- InfoGain AttributeEval output (InfoGain Rank Order, InfoGain Value, Original Row from input)

Transcripts were assigned to a list group of “DAM-InfoGain_p-value” (green) if shared between the p-value and InfoGain list (n = 149 of 1751), “DAM-p-value” (pink) if unique (i.e., not shared with InfoGain list) to p-value </= 0.05 list (n = 1,494 of 1,751); or “DAM-InfoGain” (blue) if unique to InfoGain list (n = 108 of 1,751).

9a.Sub-Folder Name: SB-WEKA_DAM_TopAttributes-InfoGain

File Name # Files Description

SB-WEKA-DAM_257-ranked.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the DAM comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 257 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA whereby bottom-ranked (of the 257) attributes were removed iteratively from the input data from the inclusion of all 257 to only the top-ranked attribute.

SB-WEKA-DAM_Top-1.csv 1

SB-WEKA-DAM_Top-2.csv 1

SB-WEKA-DAM_Top-3.csv 1

SB-WEKA-DAM_Top-4.csv 1

SB-WEKA-DAM_Top-5.csv 1

SB-WEKA-DAM_Top-6.csv 1

SB-WEKA-DAM_Top-7.csv 1

SB-WEKA-DAM_Top-8.csv 1

SB-WEKA-DAM_Top-9.csv 1

SB-WEKA-DAM_Top-10.csv 1

SB-WEKA-DAM_Top-15.csv 1

SB-WEKA-DAM_Top-20.csv 1

SB-WEKA-DAM_Top-25.csv 1

SB-WEKA-DAM_Top-50.csv 1

SB-WEKA-DAM_Top-75.csv 1

SB-WEKA-DAM_Top-100.csv 1

SB-WEKA-DAM_Top-150.csv 1

SB-WEKA-DAM_Top-200.csv 1

SB-WEKA-DAM_Top-250.csv 1

9b. Sub-Folder Name: SB-WEKA_DAM_MinusTopAttributes-InfoGain

File Name # Files Description

SB-WEKA-DAM_Minus-Top-1.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the DAM comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 257 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA whereby top-ranked (of the 257) attributes were removed iteratively from the input data from the inclusion of all 257 to only the bottom-ranked attributes.

SB-WEKA-DAM_Minus-Top-2.csv 1

SB-WEKA-DAM_Minus-Top-3.csv 1

SB-WEKA-DAM_Minus-Top-4.csv 1

SB-WEKA-DAM_Minus-Top-5.csv 1

SB-WEKA-DAM_Minus-Top-6.csv 1

SB-WEKA-DAM_Minus-Top-7.csv 1

SB-WEKA-DAM_Minus-Top-8.csv 1

SB-WEKA-DAM_Minus-Top-9.csv 1

SB-WEKA-DAM_Minus-Top-10.csv 1

SB-WEKA-DAM_Minus-Top-15.csv 1

SB-WEKA-DAM_Minus-Top-20.csv 1

SB-WEKA-DAM_Minus-Top-25.csv 1

SB-WEKA-DAM_Minus-Top-50.csv 1

SB-WEKA-DAM_Minus-Top-75.csv 1

SB-WEKA-DAM_Minus-Top-100.csv 1

SB-WEKA-DAM_Minus-Top-150.csv 1

SB-WEKA-DAM_Minus-Top-200.csv 1

SB-WEKA-DAM_Minus-Top-250.csv 1

9c. Sub-Folder Name: SB-WEKA_DAM_NegativeControl

File Name # Files Description

SB-RNASeq-DAM_150-Optimal-NegCtrl.xlsx 1 The template .XLSX file for the DAM comparison (A, B) negative control. The optimal dataset (determined based on over- and underfitting data reduction) of the top 150 attributes is included on the sheet “Optimal 150” and these values are on the sheet entitled “Transposed”. There are 55 samples designated to Dam A and 17 to Dam B per genotyping performed in year 3.

SB-WEKA-DAM_NegCtrl-1.csv 1 A .CSV file containing negative control run number 1 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-2.csv 1 A .CSV file containing negative control run number 2 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-3.csv 1 A .CSV file containing negative control run number 3 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-4.csv 1 A .CSV file containing negative control run number 4 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-5.csv 1 A .CSV file containing negative control run number 5 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-6.csv 1 A .CSV file containing negative control run number 6 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-7.csv 1 A .CSV file containing negative control run number 7 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-8.csv 1 A .CSV file containing negative control run number 8 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-9.csv 1 A .CSV file containing negative control run number 9 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-DAM_NegCtrl-10.csv 1 A .CSV file containing negative control run number 10 of 10. The RAND() function was used in Excel to randomize the sample designations.

10. Folder Name: SB-WEKA-Analysis_SIRE

File Name # Files Description

SB-WEKA_Output-SIRE.docx 1 A .DOCX file containing the copied-and-pasted summary output from each algorithm run in WEKA using data associated with SIRE (# 1-12) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists).

SB-WEKA_Results-SIRE.xlsx 1 A .XLSX file containing the copied-and-pasted percent correct classification, kappa statistic, and AUROC output from each algorithm run in WEKA using data associated with SIRE (# 1-12) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists). InfoGain AttributeEval results and figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-WEKA-SIRE_Results-SUMMARY.docx 1 A .DOCX file summarizing the percent correct classification output from each algorithm run in WEKA using data associated with SIRE (# 1-12) identifiers (attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset). Figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-RNASeq-SIRE_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and SIRE identifiers (#1-12).

SB-WEKA-SIRE_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and SIRE identifiers (#1-12) in WEKA-input format (identifiers are in the columns and transcript IDs are row-headers). This file was transposed in R Studio, as the number of columns necessary exceeds the capacity of Microsoft Excel.

SB-WEKA-SIRE_32018.arff

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and SIRE identifiers (#1-12) in WEKA format (ARFF files). (identifiers are in the columns and transcript IDs are row-headers).

SB-SIRE_InfoGain_Results.xlsx The results of the Information Gain value (“score”) assignment of all 32,018 attributes for the SIRE comparison performed in WEKA. Briefly, WEKA evaluates the information gained (Shannon’s Entropy) of each attribute for a given comparison, SIRE via the WEKA tool

InfoGainAttributeEval (SVM → SMO algorithm).

SB-RNASeq-SIRE_35-ranked.xlsx

1 An excel file containing only the attributes assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 35 attributes) and the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by SIRE grouping (#1-12). The rank order (1-35) for each attribute is included as the top row and was used to generate the files that only include a certain number of the top ranked attributes (see: sub-folders).

SB-RNASeq-SIRE_3604-significant.xlsx

1 An excel file containing only the attributes assigned a p-value below or equal to 0.05 (n = 3,604 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench (see: “SB-RNASeq_CLC-Stats_SIRE.xlsx” in the Folder: “SB_RNASeq-CLC-Statistical-Analyses_Output”) and the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by SIRE grouping (#1-12). The rank order (1-3,604) for each attribute is included.

SB-WEKA-SIRE_3604-significant.csv

1 A WEKA-formatted CSV file containing only the attributes assigned a p-value below or equal to 0.05 (n = 3,604 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench, the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by SIRE grouping (#1-12).

SB-RNASeq_SIRE_3609-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 3,609 unique transcripts (30 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 35 attributes), (2) assigned a p-value below or equal to 0.05 (n = 3,604 attributes), or (3) both. Samples (1-72) are identified by SIRE grouping (#1-12). Transcripts are in alphabetical order.

SB-WEKA_SIRE_3609-ranked-signif.csv 1 A WEKA-formatted CSV file containing the 3,609 unique attributes (30 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 35 attributes), (2) assigned a p-value below or equal to 0.05 (n = 3,604 attributes), or (3) both. Samples (1-72) are identified by SIRE grouping (#1-12). Transcripts are in alphabetical order.

SB-RNASeq_SIRE_30-shared-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 30 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by SIRE grouping (#1-12). Transcripts are in alphabetical order.

SB-WEKA_SIRE_30-shared-ranked-signif.csv 1 A WEKA formatted CSV file containing the RPKM transcript expression, TE, values for the 30 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by SIRE grouping (#1-12). Transcripts are in alphabetical order.

SB-SIRE_InfoGain-StatSignif_Results.xlsx 1 An excel file containing the following information on each sheet:

[p <|= 0.05] - the CLC Genomics Workbench descriptors of the transcripts that had a p-value less than or equal to 0.05 in the differential expression analysis between Sires (#1-12) and including p-value order (1-3,604 from smallest to largest).

[Ranked InfoGain] - the WEKA output for the 35 transcripts assigned a InfoGain (Shannon's Entropy) value above 0.0 for the SIRE comparison and including the rank order (1-35 from most to least informative among this set)

[Shared - InfoGain and p value] - CLC Genomics Workbench descriptors of the 30 transcripts that were assigned a p-value less than or equal to 0.05 and an InfoGain rank above 0.0 and including InfoGain rank and p-value orders.

[Unique - InfoGain] - CLC Genomics Workbench descriptors of the 5 transcripts that were assigned an InfoGain rank above 0.0, but that were not assigned a p-value less than or equal to 0.05 and including InfoGain rank orders.

[Unique - pvalue] - CLC Genomics Workbench descriptors of the 3,574 transcripts that were assigned a p-value less than or equal to 0.05, but that were not assigned an InfoGain rank above 0.0 and including p-value orders.

SIRE-FullList-Annotation-Evalues.xlsx 1 An excel file containing the following information about the 3,609 unique transcripts from the combined p-value and InfoGain-ranked lists for the SIRE (#1-12) comparison (3,639 transcripts total, 30 of which are on both lists, or are "shared"):

- Sample IDs for Individual SB 1-72
- SIRE (#1-12)
- CLC Genomics Workbench Differential Expression Analysis output (Chromosome, Region, Max group mean, Log2 fold change, fold change, p-value, FDR p-value, Bonferoni, p-value order),
- InfoGain AttributeEval output (InfoGain Rank Order, InfoGain Value, Original Row from input)

Transcripts were assigned to a list group of "SIRE-InfoGain_p-value" (highlighted in green) if shared between the p-value and InfoGain list (n = 30 of 3,609), "SIRE-p-value" (pink) if unique (i.e., not shared with InfoGain list) to p-value </= 0.05 list (n = 3,574 of 3,609); or "SIRE-InfoGain" (blue) if unique to InfoGain list (n = 5 of 3,609).

10a.Sub-Folder Name: SB-WEKA_SIRE_TopAttributes-InfoGain

File Name # Files Description

SB-WEKA-SIRE_35-ranked.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the SIRE comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 35 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA whereby bottom-ranked (of the 35) attributes were removed iteratively from the input data from the inclusion of all 35 to only the top-ranked attribute.

SB-WEKA-SIRE_Top-1.csv 1

SB-WEKA-SIRE_Top-2.csv 1

SB-WEKA-SIRE_Top-3.csv 1

SB-WEKA-SIRE_Top-4.csv 1

SB-WEKA-SIRE_Top-5.csv 1

SB-WEKA-SIRE_Top-6.csv 1

SB-WEKA-SIRE_Top-7.csv 1

SB-WEKA-SIRE_Top-8.csv 1

SB-WEKA-SIRE_Top-9.csv 1

SB-WEKA-SIRE_Top-10.csv 1

SB-WEKA-SIRE_Top-15.csv 1

SB-WEKA-SIRE_Top-20.csv 1

SB-WEKA-SIRE_Top-25.csv 1

SB-WEKA-SIRE_Top-30.csv 1

10b. Sub-Folder Name: SB-WEKA_SIRE_MinusTopAttributes-InfoGain

File Name # Files Description

SB-WEKA-SIRE_Minus-Top-1.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the SIRE comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 35 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA, whereby top-ranked (of the 35) attributes were removed iteratively from the input data from the inclusion of all 35 to only the bottom-ranked attributes.

SB-WEKA-SIRE_Minus-Top-2.csv 1

SB-WEKA-SIRE_Minus-Top-3.csv 1

SB-WEKA-SIRE_Minus-Top-4.csv 1

SB-WEKA-SIRE_Minus-Top-5.csv 1

SB-WEKA-SIRE_Minus-Top-6.csv 1

SB-WEKA-SIRE_Minus-Top-7.csv 1

SB-WEKA-SIRE_Minus-Top-8.csv 1

SB-WEKA-SIRE_Minus-Top-9.csv 1

SB-WEKA-SIRE_Minus-Top-10.csv 1

SB-WEKA-SIRE_Minus-Top-15.csv 1

SB-WEKA-SIRE_Minus-Top-20.csv 1

SB-WEKA-SIRE_Minus-Top-25.csv 1

SB-WEKA-SIRE_Minus-Top-30.csv 1

10c. Sub-Folder Name: SB-WEKA_SIRE_NegativeControl

File Name # Files Description

SB-RNASeq-SIRE_35-Optimal-NegCtrl.xlsx 1 The template .XLSX file for the SIRE comparison (#1-12) negative control. The optimal dataset (determined based on over- and underfitting data reduction) of the top 35 attributes is included on the sheet “Optimal 35” and these values are on the sheet entitled “Transposed”. There are 7 samples designated to Sire-1, 5 to Sire-2, 9 to Sire-3, 10 to Sire-4, 2 to Sire-5, 22 to Sire-6, 1 to Sire-7, 3 to Sire-8, 2 to Sire-9, 6 to Sire-10, 3 to Sire-11, and 2 to Sire-12.

SB-WEKA-SIRE_NegCtrl-1.csv 1 A .CSV file containing negative control run number 1 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-2.csv 1 A .CSV file containing negative control run number 2 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-3.csv 1 A .CSV file containing negative control run number 3 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-4.csv 1 A .CSV file containing negative control run number 4 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-5.csv 1 A .CSV file containing negative control run number 5 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-6.csv 1 A .CSV file containing negative control run number 6 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-7.csv 1 A .CSV file containing negative control run number 7 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-8.csv 1 A .CSV file containing negative control run number 8 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-9.csv 1 A .CSV file containing negative control run number 9 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE_NegCtrl-10.csv 1 A .CSV file containing negative control run number 10 of 10. The RAND() function was used in Excel to randomize the sample designations.

11. Folder Name: SB-WEKA-Analysis_SIRE-GROUP

File Name # Files Description

SB-WEKA_Output-SIRE-GROUP.docx 1 A .DOCX file containing the copied-and-pasted summary output from each algorithm run in WEKA using data associated with SIRE GROUP (Large, Small) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists).

SB-WEKA_Results-SIRE-GROUP.xlsx 1 A .XLSX file containing the copied-and-pasted percent correct classification, kappa statistic, and AUROC output from each algorithm run in WEKA using data associated with SIRE GROUP (Large, Small) identifiers (all attributes, all attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset, the statistically significant attributes, and the attributes shared between InfoGain ranked and p-value lists). InfoGain AttributeEval results and figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-WEKA-SIRE-GROUP_Results-SUMMARY.docx 1 A .DOCX file summarizing the percent correct classification output from each algorithm run in WEKA using data associated with SIRE GROUP (Large, Small) identifiers (attributes ranked via InfoGain, iterative runs with the bottom- and top-ranked attributes removed from the dataset). Figures of algorithm performance (percent correct classification of each model with each cross-validation method) are also in this file.

SB-RNASeq-SIRE-GROUP_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and SIRE GROUP identifiers (Large, Small).

SB-WEKA-SIREGROUP_32018.csv

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and SIRE GROUP identifiers (Large, Small) in WEKA-input format (identifiers are in the columns and transcript IDs are row-headers). This file was transposed in R Studio, as the number of columns necessary exceeds the capacity of Microsoft Excel.

SB-WEKA-SIREGROUP_32018.arff

1 Transcript expression (TE) values for all 32,018 transcripts measured in CLC Genomics Workbench for SB associated with RNASeq sample names and SIRE-GROUP group identifiers (Large, Small) in WEKA format (ARFF files). (identifiers are in the columns and transcript IDs are row-headers).

SB-SIRE-GROUP_InfoGain_Results.xlsx The results of the Information Gain value (“score”) assignment of all 32,018 attributes for the SIRE-GROUP comparison performed in WEKA. Briefly, WEKA evaluates the information gained (Shannon’s Entropy) of each attribute for a given comparison, SIRE-GROUP via the WEKA tool InfoGainAttributeEval (SVM → SMO algorithm).

SB-RNASeq-SIRE-GROUP_378-ranked.xlsx

1 An excel file containing only the attributes assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 378 attributes) and the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by SIRE GROUP identifiers (Large, Small). The rank order (1-378) for each attribute is included as the top row and was used to generate the files that only include a certain number of the top ranked attributes (see: sub-folders).

SB-RNASeq-SIRE-GROUP_2356-significant.xlsx

1 An excel file containing only the attributes assigned a p-value below or equal to 0.05 (n = 2,356 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench (see: “SB-RNASeq_CLC-Stats_SIRE-GROUP.xlsx” in the Folder: “SB_RNASeq-CLC-Statistical-Analyses_Output”) and the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by SIRE GROUP identifiers (Large, Small). The rank order (1-2,356) for each attribute is included.

SB-WEKA-SIRE-GROUP_2356-significant.csv

1 A WEKA-formatted CSV file containing only the attributes assigned a p-value below or equal to 0.05 (n = 2,356 attributes) via Differential Expression Analysis conducted in CLC Genomics Workbench, the associated data (RPKM transcript expression, TE, values) for each of the seventy-two samples in order of 1-72, but only identified by SIRE GROUP identifiers (Large, Small).

SB-RNASeq_SIRE-GROUP_2520-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 2,520 unique transcripts (214 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 378 attributes), (2) assigned a p-value below or equal to 0.05 (n = 2,356 attributes), or (3) both. Samples (1-72) are identified by SIRE GROUP identifiers (Large, Small). Transcripts are in alphabetical order.

SB-WEKA_SIRE-GROUP_2520-ranked-signif.csv 1 A WEKA-formatted CSV file containing the 2,520 unique attributes (214 shared between both lists) that were (1) assigned a rank above 0.0 by the InfoGainAttributeEval tool in WEKA (n = 378 attributes), (2) assigned a p-value below or equal to 0.05 (n = 2,356 attributes), or (3) both. Samples (1-72) are identified by SIRE GROUP identifiers (Large, Small). Transcripts are in alphabetical order.

SB-RNASeq_SIRE-GROUP_214-shared-ranked-signif.xlsx 1 An excel file containing the RPKM transcript expression, TE, values for the 214 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by SIRE GROUP identifiers (Large, Small). Transcripts are in alphabetical order.

SB-WEKA_SIRE-GROUP_214-shared-ranked-signif.csv 1 A WEKA formatted CSV file containing the RPKM transcript expression, TE, values for the 214 transcripts shared between the p-value </= 0.05 and InfoGain-ranked lists. Samples (1-72) are identified by SIRE GROUP identifiers (Large, Small). Transcripts are in alphabetical order.

SB-SIRE-GROUP_InfoGain-StatSignif_Results.xlsx 1 An excel file containing the following information on each sheet:

[p <|= 0.05] - the CLC Genomics Workbench descriptors of the transcripts that had a p-value less than or equal to 0.05 in the differential expression analysis between SIRE groups (Large, Small) and including p-value order (1-2,356 from smallest to largest).

[Ranked InfoGain] - the WEKA output for the 378 transcripts assigned a InfoGain (Shannon's Entropy) value above 0.0 for the SIRE comparison and including the rank order (1-378 from most to least informative among this set)

[Shared - InfoGain and p value] - CLC Genomics Workbench descriptors of the 214 transcripts that were assigned a p-value less than or equal to 0.05 and an InfoGain rank above 0.0 and including InfoGain rank and p-value orders.

[Unique - InfoGain] - CLC Genomics Workbench descriptors of the 164 transcripts that were assigned an InfoGain rank above 0.0, but that were not assigned a p-value less than or equal to 0.05 and including InfoGain rank orders.

[Unique - pvalue] - CLC Genomics Workbench descriptors of the 2,142 transcripts that were assigned a p-value less than or equal to 0.05, but that were not assigned an InfoGain rank above 0.0 and including p-value orders.

SIRE-GROUP-FullList-Annotation-Evalues.xlsx 1 An excel file containing the following information about the 2,520 unique transcripts from the combined p-value and InfoGain-ranked lists for the SIRE GROUP (Large, Small) comparison (2,734 transcripts total, 214 of which are on both lists, or are "shared"):

- Sample IDs for Individual SB 1-72
- SIRE groups (Large, Small)
- CLC Genomics Workbench Differential Expression Analysis output (Chromosome, Region, Max group mean, Log2 fold change, fold change, p-value, FDR p-value, Bonferoni, p-value order),
- InfoGain AttributeEval output (InfoGain Rank Order, InfoGain Value, Original Row from input)

Transcripts were assigned to a list group of "SIRE-GROUP-InfoGain_p-value" (highlighted in green) if shared between the p-value and InfoGain list (n = 214 of 2,520), "SIRE-GROUP-p-value" (pink) if unique (i.e., not shared with InfoGain list) to p-value </= 0.05 list (n = 2,142 of 2,520); or "SIRE-GROUP-InfoGain" (blue) if unique to InfoGain list (n = 164 of 2,520).

11a.Sub-Folder Name: SB-WEKA_SIRE-GROUP_TopAttributes-InfoGain

File Name # Files Description

SB-WEKA-SIRE-GROUP_378-ranked.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the SIRE comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 378 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA whereby bottom-ranked (of the 378) attributes were removed iteratively from the input data from the inclusion of all 378 to only the top-ranked attribute.

SB-WEKA-SIRE-GROUP_Top-1.csv 1

SB-WEKA-SIRE-GROUP_Top-2.csv 1

SB-WEKA-SIRE-GROUP_Top-3.csv 1

SB-WEKA-SIRE-GROUP_Top-4.csv 1

SB-WEKA-SIRE-GROUP_Top-5.csv 1

SB-WEKA-SIRE-GROUP_Top-6.csv 1

SB-WEKA-SIRE-GROUP_Top-7.csv 1

SB-WEKA-SIRE-GROUP_Top-8.csv 1

SB-WEKA-SIRE-GROUP_Top-9.csv 1

SB-WEKA-SIRE-GROUP_Top-10.csv 1

SB-WEKA-SIRE-GROUP_Top-15.csv 1

SB-WEKA-SIRE-GROUP_Top-20.csv 1

SB-WEKA-SIRE-GROUP_Top-25.csv 1

SB-WEKA-SIRE-GROUP_Top-50.csv 1

SB-WEKA-SIRE-GROUP_Top-75.csv 1

SB-WEKA-SIRE-GROUP_Top-100.csv 1

SB-WEKA-SIRE-GROUP_Top-150.csv 1

SB-WEKA-SIRE-GROUP_Top-200.csv 1

SB-WEKA-SIRE-GROUP_Top-250.csv 1

SB-WEKA-SIRE-GROUP_Top-300.csv 1

SB-WEKA-SIRE-GROUP_Top-350.csv 1

11b. Sub-Folder Name: SB-WEKA_SIRE-GROUP_MinusTopAttributes-InfoGain

File Name # Files Description

SB-WEKA-SIRE-GROUP_Minus-Top-1.csv 1 WEKA input files containing the transcript expression (TE) RPKM values for a given number (indicated in each file name) of top-ranked attributes (transcripts) for the SIRE comparison. These attributes were ranked using the WEKA tool InfoGainAttributeEval (SVM → SMO was used), which evaluates attributes based upon the amount of information the data for a given attribute provides the algorithm for learning and classification. A total of 378 attributes were assigned a rank above zero using this tool. These files were used as input into WEKA, whereby top-ranked (of the 378) attributes were removed iteratively from the input data from the inclusion of all 378 to only the bottom-ranked attributes.

SB-WEKA-SIRE-GROUP_Minus-Top-2.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-3.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-4.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-5.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-6.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-7.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-8.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-9.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-10.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-15.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-20.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-25.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-50.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-75.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-100.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-150.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-200.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-250.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-300.csv 1

SB-WEKA-SIRE-GROUP_Minus-Top-350.csv 1

11c. Sub-Folder Name: SB-WEKA_SIRE-GROUP_NegativeControl

File Name # Files Description

SB-RNASeq-SIRE-GROUP_15-Optimal-NegCtrl.xlsx 1 The template .XLSX file for the SIRE-GROUP comparison (Large, Small) negative control. The optimal dataset (determined based on over- and underfitting data reduction) of the top 15 attributes is included on the sheet “Optimal 15” and these values are on the sheet entitled “Transposed”. There are 27 samples designated as Large and 45 samples designated as Small in the study.

SB-WEKA-SIRE-GROUP_NegCtrl-1.csv 1 A .CSV file containing negative control run number 1 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-2.csv 1 A .CSV file containing negative control run number 2 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-3.csv 1 A .CSV file containing negative control run number 3 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-4.csv 1 A .CSV file containing negative control run number 4 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-5.csv 1 A .CSV file containing negative control run number 5 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-6.csv 1 A .CSV file containing negative control run number 6 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-7.csv 1 A .CSV file containing negative control run number 7 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-8.csv 1 A .CSV file containing negative control run number 8 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-9.csv 1 A .CSV file containing negative control run number 9 of 10. The RAND() function was used in Excel to randomize the sample designations.

SB-WEKA-SIRE-GROUP_NegCtrl-10.csv 1 A .CSV file containing negative control run number 10 of 10. The RAND() function was used in Excel to randomize the sample designations.

11. Folder Name: HSB-IPA-Analysis

File Name # Files Description

HSB_IPA-Canonical-Pathways_FFAR-Nov-2021.xls 1 Output from QIAGEN Ingenuity Pathway Analysis of hybrid striped bass from Objective 1, Hybrid Striped Bass Study 2. Tox Lists are lists of molecules that are known to be involved in a particular type of toxicity. These lists are hand curated by our content scientists. IPA-Tox analysis uses Toxicity Functions in combination with Toxicity Lists to link experimental data to clinical pathology endpoints, understand pharmacological response, and support mechanism of action and mechanism of toxicity hypothesis generation.

HSB_IPA-Diseases-Functions_FFAR-Nov-2021.xls 1

HSB_IPA-Diseases-Functions-Tox_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-Cardiac-Fibrosis_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-Cardiac-Hypertrophy_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-Decr-Depolarization_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-FXR-RXR-Activation_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-Hepatic-Fibrosis_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-LXR-RXR-Activation_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-Neg-PhaseResp-Proteins_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-Pos-PhaseResp-Proteins_FFAR-Nov-2021.xls 1

HSB_IPA-Lists-Reversible-Glomerulon_FFAR-Nov-2021.xls 1

HSB_IPA-Upstream-Analysis_FFAR-Nov-2021.xls 1

HSB-Canonical-Pathways_FFAR-Nov-2021.jpg 1

HSB-IPA-Lists- Hepatic-Fibrosis_Nov-2021.pdf 1

HSB-IPA-Lists-Cardiac-Fibrosis_Nov-2021.pdf 1

HSB-IPA-Lists-Cardiac-Hypertrophy_Nov-2021.pdf 1

HSB-IPA-Lists-Decr-Depolarization_Nov-2021.pdf 1

HSB-IPA-Lists-FXR-RXR-Activation_Nov-2021.pdf 1

HSB-IPA-Lists-LXR-RXR-Activation_Nov-2021.pdf 1

HSB-IPA-Lists-Neg-PhaseRes-Proteins_Nov-2021.pdf 1

HSB-IPA-Lists-Pos-PhaseRes-Proteins_Nov-2021.pdf 1

4.2. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Upload copies of records and a simple file inventory, if necessary, in a compressed folder.*

Are the data experimental, observational, simulation, or compiled/derived? Check all categories that apply.

Responses Selected:

Experimental: Data captured through a controlled experiment where an intervention is being tested. Results should be replicable and reproducible (e.g., lab experiments, field trials)

For each checked box, please provide a brief description of the data including where the data are collected / sourced.

Gene sequence data and gene expression data (RNASeq) are provided as supplementary files with this report. Additional metadata, including fish growth performance metrics also are provided. The data were sourced from fish grown at the NC State University Pamlico Aquaculture field Laboratory in Aurora, NC and the Grinnells Aquatic Research Facility in Raleigh, NC. Where appropriate, data have been submitted to the US NIH NCBI (GenBank) for archiving and public availability.

Describe (what, where, when) and list additional data (not captured in question above), plus samples, physical collections, software, curriculum materials and other materials produced during this period. Who is the expert or person(s) tasked with data collection (please provide their name and affiliation with the project)? This can be any member of the team. For data-focused projects, that individual is expected to be key personnel.

Benjamin J. Reading (PI, Associate Professor and University Faculty Scholar, NC State University): AEC 510 Machine Learning Applications in Biological Sciences course materials and curriculum; Coordinator of the US Hybrid Striped Bass Breeding Program (germplasm and domesticated fish/seedstocks); project oversight for all data collected.

Linnea K. Andersen (PhD Student, NC State University): Gene expression (RNASeq) data for striped bass and hybrid striped bass (including sample metadata of growth metrics); also design of machine learning pipelines used for analysis of data with Orion Integrated Biosciences.

Erin E. Ducharme (MS Student, NC State University): Metabolomic (small molecule profiling) data for striped bass and hybrid striped bass (including sample metadata of growth metrics).

Sarah Rajab (PhD Student, NC State University): Gene expression (RNASeq) and metabolomic (small molecule profiling) data for hybrid striped bass (including sample metadata of growth metrics).

Michael O. Frinsko (NC Cooperative Extension, Area Extension Agent Aquaculture; PhD Student, NC State University): Hybrid striped bass and striped bass growth data and larval rearing information.

Provide how the data are organized and managed for all data created and utilized during this project. This could include the relevant data standardization, data formats, data definitions (e.g., data dictionary, ontologies), data organization, and/or where the data is stored. Please note if any of these data are confidential, and if so, when you would anticipate being able to share them publicly in the future.

A file attached to this report provides detailed information (46 pages long; FFAR_Report_File_Inventory_Year5-EXTENSION_2021.docx) regarding the data files submitted for this project (terabytes in size). Where appropriate, all raw sequence data have been deposited in the US NIH NCBI (GenBank) for storage and public access (e.g., the entire genome of striped bass has been annotated and made available along with all of the source sequence data). Other raw data (e.g., growth metics) are stored on personal hard drives and will be reported in published manuscripts anticipated in 2022 (in addition to being provided in this report).

FFAR supports the FAIR principles. We would like to understand how your project aligns with FAIR principles (for definition, see Go Fair, for more information see Nature Magazine). Does your data align with the 4 principles in FAIR? Please check all boxes that apply and provide a brief description:

Responses Selected:

Findable – how could your data be found (e.g., online searches, publications)?
Access - how will the data be accessed and shared by researchers and the general public, respectively (e.g., is the data publicly accessible, proprietary, temporarily proprietary)?
Interoperable – are the data connected to other sources (e.g., do APIs currently exist, ontologies / standards that enable easy connection between data)
Reusability - describe the policies and provisions for re-use, re-distribution, and the production of derivatives.

For each checked box, please provide a brief description.

As above and as detailed in the report, the data storage adheres to all four of these principles. For example see: https://www.ncbi.nlm.nih.gov/assembly/GCF_004916995.1/?&utm_source=gquery
This is one online link to data generated from this project that are archived at the US NIH NCBI and made publicly available through the internet. All of the data stored at NCBI adhere to these FAIR guidelines (the information is required when uploading the data); all other data archived at NCBI are provided elsewhere in this report (with links) and are publicly available.

What are the provisions for the data for the appropriate protection of privacy, confidentiality, security, intellectual property, or other rights or requirements?

The data are not proprietary nor do they contain sensitive information.

Describe the plans for archiving data, samples, and other research products, and for preservation of access to them. Refer to the FFAR Data Guidance Document for more information.

See above for data storage. Replicate samples collected for this project are stored at -80 degrees C (ultra cold) at NC State University. Importantly, the genetic stocks of domesticated striped bass and white bass (germline / live animals / broodstocks) are maintained at NC State University and also backup of these live broodstock animals are maintained at USDA ARS Stuttgart National Aquaculture Research Center (Stuttgart, AR). These broodstocks are provided to farmers who with to raise the animals as described in the report.

Will this project coordinate with existing FFAR-supported grants and their data products? If so, provide a brief description of how, including the project title and PI of the existing FFAR work If not, provide a brief rationale for why this project does not coordinate with existing FFAR-supported work.

Yes:

PI: Benjamin Reading

Title: The Hybrid Striped Bass: Understanding Heterosis to Improve Food-Animal Ag

We provide a novel pipeline for analysis of "Omic" data using machine learning. The approach can be used to evaluate gene expression data as well as metabolomic data to predict a phenotype. Overall, the pipeline provides a way to analyze allelic gene expression as required to evaluate heterosis--not only are we measuring the expression of a particular gene, but we can evaluate whether it is the maternal or paternal gene (or both) being expressed in the hybrid. To our knowledge this has not been achieved before and we would be glad to work with others on different animal and plant systems as well as with different forms of Omics data to further ground truth this application. This may lead to a novel way of analyzing Omics data in 1) a "true" systems biology approach and 2) predictive phenomics. We also wish to further evaluate the pipeline and standardize it for universal applications with Omics data, create a software package, and otherwise inform the scientific community that approaches such as this are the 4th Industrial Revolution of 21st Century Biology/Agriculture.

Please provide any feedback on these new data questions including any areas where you would like to see more clarity or explanation.

For projects with large amounts of data (such as this one--few to many TB) we already provided a detailed description of the data files...it is WAY more than can fit into one of these small windows! As above, our description of the data file inventory is 46 pages long. If there is a problem understanding our data manifest, please let us know.

Certification

When completing tasks on SM Apply, please note that only the owner of the application (i.e., the user whose name is marked ‘owner’ in parentheses on the menu on the left of the screen when you open your award) can hit ‘Submit’ on finished tasks. If you are not the owner, the ‘Submit’ button will not appear. FFAR does not receive notification of completed tasks without a submission.

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5. CERTIFICATION

I hereby certify that to the best of my knowledge and belief, the above report and all supporting documents are true, accurate, and complete. I am aware there is a significant penalty for submitting false or misleading information.

By checking the box below, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name*

Dr. Benjamin Reading

PI Electronic Signature*

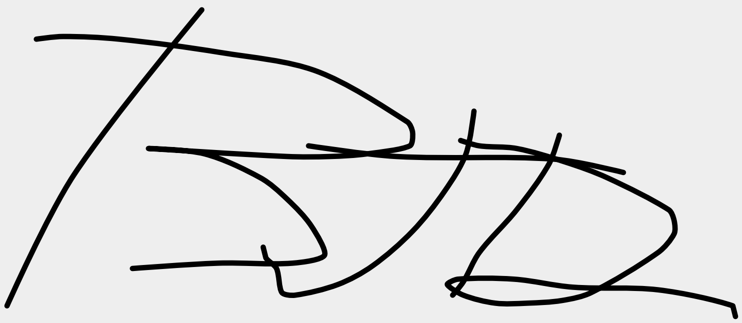
Date*

12/15/2021

Authorized Signing Official (ASO)*

Ms. Sherrie Settle

ASO Electronic Signature*



Date*

12/15/2021